Separate Ways and the Upside Down: the classical world and the field equations from a CPT fold, and the neutrino line that would break it

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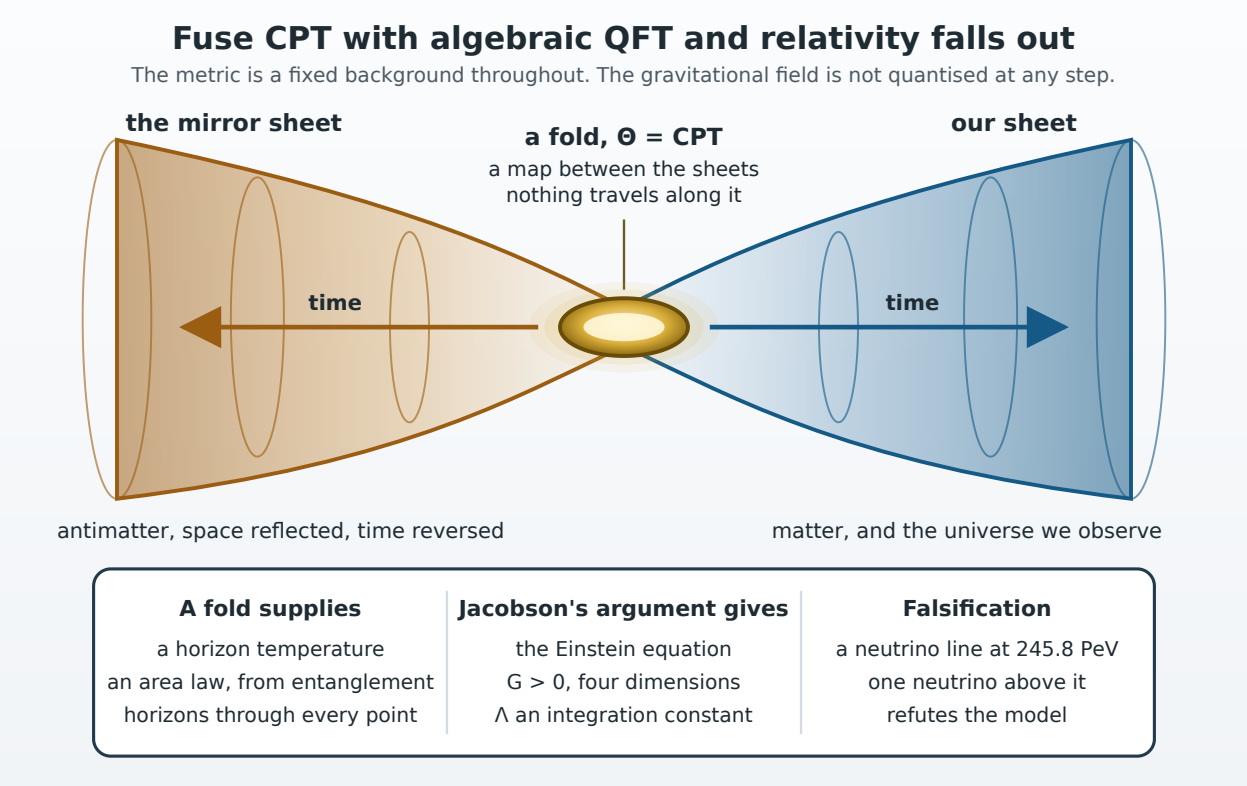
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Abstract

We study what follows if CPT is a symmetry of the universe itself and not only of its laws, as Boyle, Finn and Turok proposed: the big bang is then a fold, with a mirror-image sheet on its far side. We combine it with a standard result of algebraic quantum field theory, that the modular conjugation of a wedge is a reflection. Because the fold is a reflection, every field splits into the average of its two copies and their difference, and these behave as the classical and the quantum variable. At a horizon the fold relates the two sides by half a thermal period, so the state across it is a thermofield double, pure as a whole and thermal to each side. At a flat wedge the fold's map is the CPT operator, and the Bisognano-Wichmann theorem that says so also makes the vacuum thermal on every wedge. That thermal state is where the Unruh temperature and the entanglement area law come from, the two inputs Jacobson's 1995 derivation of Einstein's equations took from quantum field theory. We re-derive his argument from the fold. The gravitational field is not quantised at any step. Nineteen properties of general relativity follow from four inputs, two that define a metric theory and two measured constants. Nine were already in Jacobson's argument and four are standard; six need the folded universe, among them four dimensions and the classical-quantum split. Applied at the bang, the same construction sets a minimum on particle production, and the observed dark-matter abundance turns it into predictions that can fail: a dark-matter fermion of at most 491.6 ± 2.0 PeV, a neutrino line from its decay at 245.8 ± 1.0 PeV, and a neutrino-mass sum of at least 58.8 meV, which the model cannot relax because it keeps dark energy constant. The model departs from general relativity only inside black holes, in a shell 2.1 microns thick at one solar mass, where relativity predicts its own breakdown.

1. Introduction



Graphical abstract. One postulate goes in: CPT is a symmetry of the universe itself, not just of the laws inside it. The fold relating the two sheets is an involution, a relation between the sheets with nothing travelling along it, so no matter crosses it and no white-hole population or traversable connection is predicted. The classical-quantum split is §2.1, the horizon temperature §3.1, the field equations §3.6, and the count line by line Appendix E.

Boyle, Finn and Turok proposed that CPT, the combination of charge conjugation, spatial reflection and reversal of time orientation, is a symmetry of the universe itself and not only of its laws [9,10,11]. The Big Bang then has a far side, our own universe over again with matter swapped for antimatter, space reflected and time running the other way. This paper adds one thing to their proposal, a result from algebraic quantum field theory that fixes the map between the two sides, and follows what comes out. What comes out includes Einstein's field equations, by Jacobson's thermodynamic route with the metric a fixed background throughout, and an upper bound on the mass of their dark-matter particle. The map relating the two sheets undoes itself when applied twice, which makes it an involution. Throughout this paper *the fold* names that involution, and also the two-sheeted universe built on it.

Their cosmology needs no inflaton. Two sheets meet at a radiation bang, CPT relates them, and the dark matter is a heavy right-handed neutrino produced gravitationally at the bang, its mass fixed by the observed abundance. Two things in it are left open. The abundance calculation has to be given a quantum state to start from, and the cosmology selects one by a minimum-energy condition brought in from outside. The cosmology also says that some involution relates the sheets without saying which.

A second body of work, developed independently, answers both questions. Algebraic quantum field theory, AQFT, describes a field by the algebra of observables in each region, and attaches to such an algebra a mirror map called its modular conjugation. For the vacuum algebra of a static patch the map is known: Sewell's theorem [2] and the de Sitter analysis of Borchers and Buchholz [3] identify it with a reflection through the horizon, the wedge reflection, and Bisognano and Wichmann [28] supply the flat-space original. Parikh, Savonije and Verlinde [1] show by parallel transport between antipodal points that the map acts on the tangent space by PT. Chang and Li [26] give the mode-level form and the reality condition.

Three older tools are used with it. Thermofield dynamics, due to Takahashi and Umezawa [29] and applied at horizons by Israel and Maldacena [7,8], writes a thermal state as a pure state of two copies of a system. The closed time path of Schwinger and Keldysh [16,17] calculates expectation values by running time forward and then back along a second leg. The symmetric thermal contour of Niemi and Semenoff [18] and Herzog and Son [19] puts that second leg half a thermal period away. None of these ingredients is new here.

This paper joins the two bodies of work, and each fills a gap in the other. The wedge-reflection theorems fix the involution the cosmology leaves undetermined. That involution, applied at the bang, supplies the family of states the cosmology had to prescribe. The cosmology picked one state from the family and got one value for the dark-matter mass; an operator inequality over the whole family turns the observed abundance into an upper bound on it. Boyle, Finn and Turok reach the state that saturates the bound by an independent minimum-energy route, so §2.3 does not have to take it on their authority.

Two frameworks can fit together formally without the fit meaning anything, so the useful question is what the combination predicts. The dark-matter results came first, and the gravitational results came out of stress-testing them. General relativity is a demanding test. If CPT read through the modular structure of a wedge is physics, the field equations should follow without being put in.

Three things often expected of a quantum theory of gravity follow with the metric held fixed as a background. First comes the divide between classical and quantum, which is whether a field is even or odd under the fold and has nothing to do with scale. The second is a horizon temperature: the fold relates the two sides of a horizon by half a thermal period, and that shift makes the state across it a thermofield double, a state that is pure taken whole and thermal to either side alone. Quantum field theory on a fixed background has given that temperature since Hawking and Unruh [42,37]; what the fold adds is that it comes from the same map as the first. Third is the Einstein equation.

Jacobson showed in 1995 that the Clausius relation, applied to the local horizons an accelerated observer sees, gives the field equations [33]. He took the horizon temperature from the Unruh effect [37], the entropy proportional to area from the entanglement of vacuum fluctuations across the horizon that Bombelli, Koul, Lee and Sorkin had computed [38], and the horizons themselves from local flatness, and he argued that an equation of state need not be quantised. A line of work followed that obtains the field equations from horizon thermodynamics and entanglement [45,46,40,47]. What the fold adds is where Jacobson's inputs come from. At a flat wedge its map is the CPT operator: Bisognano and Wichmann showed that the modular conjugation of a wedge is CPT composed with a rotation by π in the transverse plane [28], and the transverse antipode is that rotation. The same theorem makes the vacuum thermal on the wedge, and the temperature and the area law are read off that thermal state. So the involution that defines the folded cosmology is the one behind Jacobson's two inputs. §3.6 re-derives his argument from it, with nothing gravitational quantised and with Newton's constant and the cosmological constant left as measured inputs, and §4.1 checks the dependency graph for any use of general relativity in its own derivation.

§3.6 works the derivation through. Appendix E lists what comes out and what goes in, nineteen properties of general relativity from four inputs. Nine of them were already in Jacobson's argument and four are standard results; six need the folded universe, and the fold also adds one non-local term to the field equations. A script produces the list so the count can be checked.

The dark-matter results are what make the model falsifiable, and most of the arithmetic goes there. At a radiation bang the equation for each field mode is an avoided crossing, the two-level problem a solid-state physicist meets when levels sweep past one another. A state the fold leaves unchanged must sit half in each component at the crossing, which leaves one free phase per mode. For every such state, pure or mixed, Gaussian or not, the average of the particle numbers before and after the crossing is bounded below on each pair of modes. A given abundance made of more particles means a lighter particle, so a floor on production is a ceiling on mass. The abundance match therefore returns

$$M_1 \leq 491.6 \pm 2.0 \text{ PeV}, \qquad E_\nu \leq 245.8 \pm 1.0 \text{ PeV}, \qquad \Sigma m_\nu \geq 58.8 \text{ meV}.$$

First is a ceiling on the dark-matter mass over every state the fold permits. The second is the energy of the neutrino from the two-body decay in the model of §2.4, half the mass, and the line is narrow enough

that one securely assigned event above it refutes the model. Third is a floor on the sum of the neutrino masses. A model usually relaxes such a floor by letting dark energy vary in time, the $w_0 w_a$ freedom; the fold is committed to constant dark energy and cannot. The quoted widths propagate the measured inputs only, at fixed production history and particle content; §2.3 sets out what is included and what is left out.

The same involution can be carried to a black-hole horizon, and that work is in the companion paper [32]. The results here do not need it, but §3.6 cites it for two things: that the surfaces where the sheets touch are caustics, which is what lets the focusing argument of §3.6 act on them, and that the sheets touch nowhere outside a horizon and only inside $r = M$ within one.

Section 2 gives the fold, the crossing and the abundance machinery, and section 3 the results and their tests. Section 4 sets out what the fold claims, its limits, and what would refute it, and section 5 concludes. The appendices carry the algebra, the states at the bang, the quotient readings, the benchmarks, and the list of what goes in and what comes out.

2. Construction of the fold

A note on method comes first. Every number quoted in this paper is produced by a script, and each script is named for the passage it supports. The calculations are in R 4.5.2 with no package loaded, so base R runs all of them, and in Python 3.12.13 where a special function or a symbolic step is wanted, using NumPy 2.5.3, SciPy 1.18.1, SymPy 1.14.0 and mpmath 1.3.0. A checking pass runs over both manuscripts on every change and fails if a number in the text and the script named for it come apart; the code and that pass are at https://github.com/BenWiseman/separate-ways.

2.1 Fold parity, and the classical-quantum split it makes

Let α be the antipodal map of de Sitter space, which sends each point to the one opposite it. It is an involution and it is free, meaning it leaves no point where it was. In the embedding of de Sitter in $\mathbb{R}^{1,4}$ it is total inversion, $X \mapsto -X$. Lifted antilinearly, so that the lift conjugates the numbers as it carries a field across, total inversion is the PCT prescription itself (PCT is Pauli's letter order for the CPT theorem), and Chang and Li use exactly that to establish CPT invariance of the scalar theory on elliptic de Sitter [26]. So the operator relating the two copies exists by the PCT theorem and the geometry. Call it Θ. What is hypothesised is narrower: that the image under that operator is the second leg of the closed time path, the return leg along which the formalism runs time backwards. The same operator can be used in three different ways. One can quotient spacetime by it, as the one-copy readings do and the companion tests; impose a reality condition on the quotient, as Chang and Li do; or keep both sheets and use the operator to pair the contour legs, which is what follows.

Where the second leg sits is not a further choice. For a state at inverse temperature β, the real-time thermal formalism allows it anywhere in a one-parameter family, displaced downward in imaginary time by any σ between 0 and β, and the physical in-in correlators come out the same for every member, which is why the displacement is usually treated as a convention. The fold cannot treat it as one. Θ is antilinear with $\Theta^2 = 1$, so exchanging the sheets is a symmetry of the pair and the cross-sheet kernel has to come out the same either way round. A contour carries two cross kernels, one per ordering of the legs. With W the two-point function of a single mode of frequency ω, their difference is

$$W(-t - i\sigma) - W(t - i\sigma) = \frac{i \sin \omega t \, \sinh(\omega(\beta/2 - \sigma))}{\omega \sinh(\beta\omega/2)},$$

checked to machine precision against the mode sum with the Bose factor put in by hand. It vanishes for all t at exactly one σ, and that is $\beta/2$. The two ends of the family are the extremes. At $\sigma = 0$ the orderings differ by the whole commutator $i \sin \omega t/\omega$, the same at any temperature, and at $\beta/2$ they do not differ at all. So the hypothesis is only that the legs are paired. An involution can pair one contour out of the family, and the symmetric one is that contour. The selecting is done by the Kubo-Martin-Schwinger (KMS) condition,

which is what being thermal means for a state. It enters through $(1+n)e^{-\omega\sigma} = ne^{\omega\sigma}$, where n is the Bose factor, and with the Bose factor removed no σ is preferred at all. Every mode is sensitive to the displacement, most sharply those with $\beta\omega$ of order two.

Write J for the point map that the wedge reflection of Sewell [2] and Borchers and Buchholz [3] implements; J here always means that map of points, and never the operator on states that implements it. In the mode convention used here

$$\alpha = J \circ P_\perp, \qquad P_\perp Y_{\ell m} = (-1)^\ell Y_{\ell m}.$$

The bifurcation sphere is the two-sphere where the future and past horizons cross. $P_\perp$ reverses orientation on it and lies outside the connected de Sitter group, so no rotation absorbs it; J fixes that sphere pointwise while α is free. That difference is the whole of what is proposed. On de Sitter the embedding fixes α, and at a black hole the geometry fixes it too. A free involutive isometry of a round bifurcation sphere is an element of $O(3)$ with no $+1$ eigenvalue, which leaves $-\mathrm{Id}$ and nothing else: a numerical search that builds involutive isometries from random orthogonal frames finds 2506 free ones, and every one is $-\mathrm{Id}$ exactly.

Kerr's surface is not round. Its dependence on θ runs through $\cos^2\theta$ alone, and no two latitudes in the half range share both a Gaussian curvature and a circumference at any spin up to $a/M = 0.998$, so its isometry group is exactly $O(2) \times \mathbb{Z}_2$; of that group's eight involutions exactly one has no fixed point, $\theta \mapsto \pi - \theta$ with $\phi \mapsto \phi + \pi$. Those two cases exhaust the stationary vacuum holes in four dimensions. Charge moves r_+ and nothing the argument uses, so the count is one at every charge admitting a horizon, and on the round S^{D-2} of a higher-dimensional hole the eigenvalue argument runs unchanged, so it is one at every dimension as well. $P_\perp$ is a choice at none of them. Where it fails is a bifurcation surface with no equatorial symmetry, and there no such map exists (companion, A.10).

The one-copy readings take spacetime to be the quotient and implement α linearly on fields. The fold keeps the time-orientable cover and equips its field algebra with an antilinear lift

$$\Theta i \Theta^{-1} = -i, \qquad \Theta \phi(f) \Theta^{-1} = \phi(f \circ \alpha).$$

No trajectory through spacetime is implied. Those two readings need not share admissible states, global observables or response laws, which is why the companion's tests name the state or response they test.

Write the two fields as Φ and $\Theta\Phi$ and define the Keldysh variables

$$\Phi_c = \frac{\Phi + \Theta\Phi}{2}, \qquad \Phi_q = \Phi - \Theta\Phi.$$

For real test functions supported in one static patch, with the antipodal partner causally disjoint so that $\Delta(f, g \circ \alpha) = 0$, time reversal changes the sign of the causal kernel while antilinearity changes the sign of i. The same-sheet commutators have opposite signs, the cross-sheet commutators vanish, and

$$[\Phi_c, \Phi_c] = [\Phi_q, \Phi_q] = 0, \qquad [\Phi_c, \Phi_q] = i\Delta,$$

the standard closed-time-path algebra, with Δ the causal commutator kernel. That doubled algebra is Takahashi and Umezawa's [29]: for left and right multiplication $L_A X = AX$ and $R_A X = XA$, one has $[L_A, L_B] = L_{[A,B]}$, $[R_A, R_B] = -R_{[A,B]}$ and $[L_A, R_B] = 0$, which for free fields with a central commutator give exactly the algebra above. The average has a classical Gaussian distribution, but average and difference together remain a quantum system, and keeping both retains the conjugate information that a projection onto the average alone would discard. What the fold adds is the spacetime involution, its state implementation and its compatibility with the free gravitational constraints (Appendix A).

The algebra above is standard. What the fold adds is a reading of it. Since $\Theta^2 = 1$, the average is exactly the fold-even part of the field and the difference exactly the fold-odd part. **On this reading the fold's**

parity is the classical-quantum split: where the two sheets agree the difference vanishes and the physics is classical, and where they disagree is where the commutator lives. The doubling is sixty years old and has been used as a calculating device, with the second copy not located anywhere. Here the second copy is a place, the other sheet, and that is what this paper adds to the algebra.

One consequence is that the opposite time orientation of the two sheets, which the cosmological accounts assume, can be derived. Tomita-Takesaki theory attaches to an algebra a modular operator Δ alongside the conjugation J (this Δ is not the commutator kernel above), and the powers Δ^{is} generate a flow, the modular flow. For a wedge the modular flow is the boost. In the embedding that boost is $K = X_0\partial_1 + X_1\partial_0$, so $dX_0/ds = X_1$, positive throughout the right static patch and negative throughout its antipodal partner. Relative to one fixed time orientation the two patch algebras' modular flows run oppositely, so the opposite time orientation of the two copies follows from reading Θ as the modular conjugation. Two limits apply. It is a statement about orientation and not about entropy, since an equilibrium modular flow does not grow entropy. And the relation at work is $\Delta_{A'} = \Delta_A^{-1}$ between an algebra and its commutant rather than a property of J, since $J\Delta^{is}J = \Delta^{is}$.

The sign and both limits are calculated. Take four thousand points of the right static patch and four thousand of its antipode. On the first dX_0/ds is positive at every point, on the second negative at every point. A rotation generator gives no sign at all, and points outside both patches give both signs in nearly equal numbers. The definite sign therefore belongs to the patch and to the boost, and the coordinate has no part in it. The two limits are checked in a finite-dimensional Tomita-Takesaki model in which each modular operator is built from the defining relation $S(a\Psi) = a^\dagger\Psi$ of its own algebra rather than written down: S_L implements the adjoint on the left algebra and fails on the commutant, each factors as $J\Delta^{1/2}$ with the same J, and the two modular operators compose to the identity, which gives $\Delta_{A'} = \Delta_A^{-1}$. Along the flow the entropy is constant to twelve figures, and J commutes with the flow. The identification of the two patches with the expanding and contracting branches is a separate interpretation, and §3's numbers use none of the patch algebra.

2.2 A radiation bang as an avoided crossing

At a radiation bang the Ricci scalar vanishes, $R = 0$, and so does a'', where a is the scale factor and a prime is a derivative in conformal time η. Massless free fields do not notice the expansion in such a background, so only a mass term can tell the two time orientations apart. With $a(\eta) = a_1\eta$ the conformal mass term is $ma = \gamma\eta$, $\gamma = M_1a_1$, and a fermion mode of comoving momentum p obeys

$$i\,\partial_\eta\psi = \begin{pmatrix} \gamma\eta & p \\ p & -\gamma\eta \end{pmatrix}\psi.$$

This is the Landau-Zener problem, with the momentum as the gap and γ as the sweep rate. The identification is a calculational handle and no priority is claimed for it. Direct integration checks the results below at $\gamma = 1$.

The fold on one mode. H is real and symmetric and $H(-\eta) = \sigma_x H(\eta)\sigma_x$, so the antilinear map $\Theta : \psi(\eta) \mapsto \sigma_x\psi^*(-\eta)$ preserves the equation and satisfies $\Theta^2 = 1$. Invariance as a ray requires $\sigma_x\psi^*(0) = e^{i\delta}\psi(0)$, which forces $|\psi_1(0)| = |\psi_2(0)| = 1/\sqrt{2}$, the mode is exactly half in each component at the bang. What survives is one relative phase, $\psi(0) \propto (1, e^{i\mu})/\sqrt{2}$, and since the minimising phase moves with p it is a function $\mu(p)$.

What is prior. Nadal-Gisbert, Navarro-Salas and Pla [22] give, for a fermion in a CPT-invariant radiation-dominated universe, the mode relation $h_k^I(\tau) = h_k^{II*}(-\tau)$ as the condition for a CPT-invariant vacuum (their eq. 66); the consequence $|h_k^I(0)| = |h_k^{II}(0)| = 1/\sqrt{2}$ at the bang (68); the one-parameter family $h_k^I(0) = e^{+i\Theta_k}/\sqrt{2}$, $h_k^{II}(0) = e^{-i\Theta_k}/\sqrt{2}$ with Θ_k an arbitrary angle (69); and the ultraviolet decay $\Theta_k \sim -\gamma/4k^2 + ...$ that Hadamard regularity requires of it (80). Those are the three statements just derived, reached by a different route, and we claim no priority for any of them. Boyle et al. parametrise the same family as $|\hat{\beta}_\pm(p)|^2 = [1-\cos 2\eta(p)\cos\lambda(p)]/2$, so that $n_\eta - n_0 = \cos\lambda(p)\,\sin^2\eta(p) \geq 0$ with $\cos\lambda > 0$ on the established range $-\pi/2 < \lambda(p) < \pi/2$, and state that the particle number density is minimised at $\eta(p) = 0$, where their state has both minimum expected particle density and minimum energy density [10].

Figure 1 draws the crossing and what the second sheet does to it.

Two sheets complete the crossing. The Landau-Zener closed form applies to a sweep from $\eta = -\infty$ to $+\infty$. A single-sheet cosmology begins at the bang and gets half of one. A full sweep returns $|\beta|^2 = e^{-\pi p^2/\gamma} = e^{-x^2}$ to a maximum relative error of 5×10^{-3} across $p \in [0.1, 1.4]$, the "in" vacuum of Appendix B.1. The sweep from $\eta = 0$ returns the $\gamma^2/16p^4$ tail of the bang-adiabatic state, with ratio 1.1145 at $p = 2$ and log-slope -4.305 against the independently derived 1.114 and -4.308. The second of these is the bang-adiabatic state, the one obtained by starting each mode in its adiabatic vacuum at the bang itself. Θ-invariance alone does not separate the two, because the bang-adiabatic state is itself in the family, at $\mu = \pi$. What separates them is the energy integral, which diverges logarithmically at $\mu = \pi$ and converges by $x \simeq 3$ at μ_*. So the fold supplies the family, and regularity excludes a member.

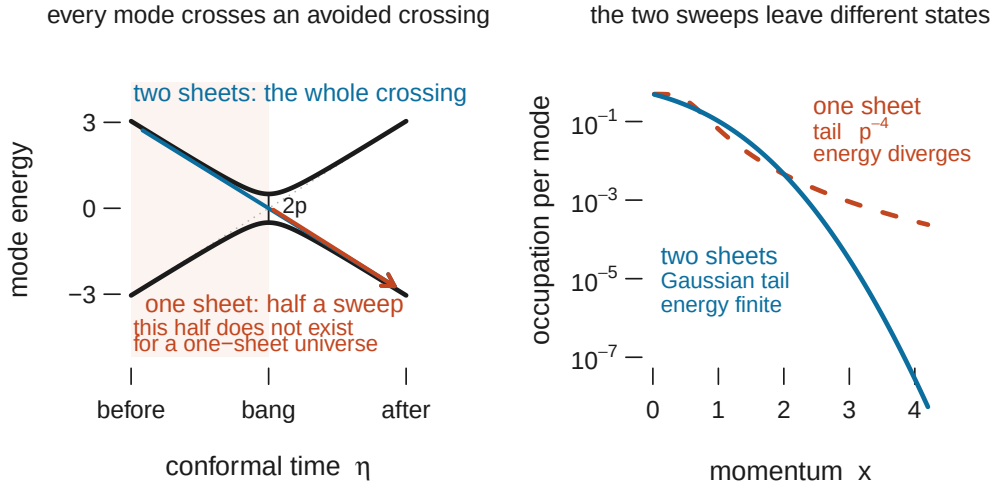


Figure 1. Every mode at a radiation bang passes through an avoided crossing; the gap at closest approach is $2p$. A universe with one sheet begins at the bang and gets half the sweep. A universe with two runs the whole way through. The right panel is what each leaves behind: the half sweep produces a p^{-4} occupation tail whose energy integral diverges logarithmically, and the whole crossing produces the Gaussian the observed abundance needs. Both curves are the closed forms, and the one-sheet curve is drawn as an illustrative profile with the stated tail index rather than a particular state.

The adopted occupation. In the dimensionless momentum $x = \sqrt{\pi}\, p/\sqrt{\gamma}$ the least-occupied member of the family is

$$n(x) = \frac{1 - \sqrt{1 - e^{-x^2}}}{2},$$

Boyle, Finn and Turok's half-angle state. Its production integral, and the ratio §3.5 uses, are

$$I = \frac{1}{\pi^2}\int_0^\infty x^2 n(x)\, dx = 0.0127597, \qquad R = \frac{\int_0^\infty x^2[-\log(1 - n(x))]\, dx}{\int_0^\infty x^2 n(x)\, dx} = 1.07037.$$

The occupation function and I come from Boyle et al. [10,11]; the release retains $I = 0.0127596673634$ as a quadrature check on the fixed occupation function.

2.3 From abundance to mass

Their small-Weyl-coupling abundance branch gives $M_1 \propto \rho_{\mathrm{DM},0}^{2/5} s_0^{-2/5} I^{-2/5} \hat{\mu}^{3/5}$. Here $\rho_{\mathrm{DM},0}$ is the present dark-matter density, s_0 the present entropy density, $\hat{\mu} = (4\pi G/3)^{-1/2}$ a Planck-scale mass and g_* the number

of relativistic species at production. The reconstruction uses $\rho_{\mathrm{DM},0} = 9.7 \times 10^{-48}$ GeV^4, $s_0 = 2.2215 \times 10^{-38}$ GeV^3, $\hat{\mu} = 5.966 \times 10^{18}$ GeV and $g_* = 106.75$, treated as exact, and returns

$$I \simeq 0.01276, \qquad M_1 = 4.916 \times 10^8 \text{ GeV}.$$

The entropy density needs one remark. A value of 2.3×10^{-38} GeV^3 is often quoted and is high by 3.5 per cent; $s_0 = 2891.2$ cm^{-3} is 2.2215×10^{-38} GeV^3, and since $M_1 \propto s_0^{-2/5}$ the benchmark with the higher value is 484.8 PeV and the half-mass energy 242.4 PeV, a shift of 6.8 PeV against a propagated width of 2.0. Those digits record how the arithmetic follows from the inputs; the physical scales are about 490 and 245 PeV. State, radiation history and branch remain inputs, and the separate large-Weyl-coupling branch changes the mass.

That width propagates the measured inputs. The abundance enters at the two-fifths power and is known to one per cent from $\Omega_{\mathrm{DM}}h^2 = 0.1200 \pm 0.0012$, contributing 0.40 per cent; the entropy density is fixed by the cosmic microwave background temperature T_{CMB} to 0.07 per cent and contributes 0.026; the production integral is a quadrature of a fixed function and the reduced Planck mass a recommended constant of the Committee on Data of the International Science Council, both negligible. Nearly all the variance comes from the abundance, giving ± 2.0 PeV on the mass and ± 1.0 PeV on the half-mass energy. It contains no allowance for the history being different or for the production model being wrong, and either moves the endpoint by far more than 2 PeV. A per cent of theory error anywhere in the reconstruction moves the endpoint by about 2 PeV on its own. The Landau-Zener treatment, $g_* = 106.75$, the choice of branch and the radiation history each shift the number without widening it, and that is the list. State freedom is different: a different admissible state changes the abundance-matched mass below the ceiling (§3.1) and does not move the ceiling.

2.4 Decay model

We adopt Boyle, Finn and Turok's particle content and their stabilising rule, a $\mathbb{Z}_2$ symmetry under which one sterile neutrino is odd and so cannot decay [10,11]. If the rule is exact the chosen sterile neutrino cannot decay and one light neutrino is massless in the stated seesaw approximation. A small added Yukawa column supplies a concrete weakly broken example with a hard two-body neutrino energy near half the heavy mass, with $h\nu : Z\nu : W\ell = 1 : 1 : 2$ at tree level; its lifetime and flavour direction are further inputs, and radiation and propagation shape any observed spectrum.

The first question about a decaying relic is why it is still present. The lifetime is an input here and it is not an unconstrained one. Appendix C.2 keeps the stabilising $\mathbb{Z}_2$ as an assumption, after showing it cannot be derived as a holonomy of the fold's topology.

An assumed global symmetry is not expected to survive quantum gravity, so if it is broken by Planck-suppressed operators and by nothing else, an operator of dimension $4 + k$ with an order-one coefficient gives $\Gamma \sim (c^2/P) M_1 (M_1/M_{\mathrm{Pl}})^{2k}$ on dimensions alone, with c the coefficient and P a phase-space factor, 1 or 16π according to convention. The mass is already fixed, so the operator dimension is the only discrete freedom left, and it is strongly selected: at $k = 3$ the estimate brackets the 10^{29} to 10^{31} s window left by external bounds, returning 1.58×10^{31} s at $P = 16\pi$ and 3.14×10^{29} s at $P = 1$, while $k = 2$ gives a particle gone long ago and $k = 4$ one that never decays, each missing under both conventions by about twenty orders. That is dimensional analysis and no more, no dimension-seven operator is written down here and none of its coefficients is computed. It also turns on the suppression scale being the ordinary Planck mass, since the reduced one puts the same estimate two orders below the window, and that choice is a convention. None of it touches the line's energy, which is kinematic. The companion checks that the induced light-neutrino mass and abundance change can be negligible. The KM3NeT event [21] is observational context, not evidence for this particle.

2.5 Tests

The endpoint test draws events from an E^{-2} spectrum above 50 PeV, truncated at the true endpoint, and smears them with a lognormal response of width 0.3 in $\ln E$. The threshold is set so that a population obeying the model fires the test with probability 0.05 whatever the sample size, and the power is the probability of firing when the true endpoint is r times the model's. A fixed per-event threshold would not do, because its sample-wide false alarm grows with the sample; that is why the calibrated threshold climbs with N in Table 2.

Assignment of an event to the decay component uses direction. A decaying halo traces the line-of-sight integral of the dark-matter density and an astrophysical population does not. The numbers assume a Navarro-Frenk-White (NFW) halo, pure Galactic decay and uniform exposure.

For the neutrino-mass sum the floor is $\sqrt{\Delta m^2_{21}} + \sqrt{\Delta m^2_{31}}$, with global-fit central values and symmetric one-sigma errors $\Delta m^2_{21} = (7.53 \pm 0.18) \times 10^{-5}$ eV^2 and $\Delta m^2_{31} = (2.510 \pm 0.030) \times 10^{-3}$ eV^2, the one set used wherever the floor's width is quoted. The comparison with the second data release of the Dark Energy Spectroscopic Instrument, DESI DR2, uses the parabolic profile likelihood Elbers et al. publish for their baryon-acoustic-oscillation and microwave-background (BAO+CMB) analysis [12], Gaussian with $\mu_0 = -0.036$ eV and $\sigma = 0.043$ eV, obtained by maximising over nuisance and cosmological parameters with degenerate masses, as a proxy likelihood under a flat non-negative-mass prior. A profile is not a marginalised posterior, matching one quantile does not reconstruct a distribution, and this is an approximation they do not make. Its warrant is that truncating at zero and renormalising returns a ninety-five per cent upper limit of 63.9 meV against the 64.2 meV they quote, a difference under half a meV. Three toy shapes matched to the published limit at the ninety-fifth percentile, an exponential from zero, a half-Gaussian at zero and a flat posterior, bracket the shape dependence. A region of practical equivalence around the floor, fixed by the floor's own propagated width before the partition is looked at, is the standard treatment of a point prediction against a bounded measurement in Bayesian psychometrics and replaces two one-sided constructions with one statement.

3. Matter sector and field equations

3.1 A ceiling over every admissible state

This section shows that the mass of §2.3 is the largest that any fold-invariant state allows. Each mode pairs a particle with an antiparticle of opposite momentum. Write a, b for the pair's operators long after the crossing, the out operators, and absorb phases so that the operators long before it, the in operators, are $a_- = c\,a + s\,b^\dagger$ and $b_- = c\,b - s\,a^\dagger$, with $c = \sqrt{1-P}$, $s = \sqrt{P}$ and $P = |\beta|^2$ the production probability of the crossing. Let $N_\pm$ be the number per pair member in each asymptotic region and $Q = (N_+ + N_-)/2$. On the four-dimensional pair block Q has eigenvalues $\{n_*, \frac{1}{2}, \frac{1}{2}, 1-n_*\}$ with $n_* = (1-\sqrt{1-P})/2$, the two middle values belonging to the odd-parity states, so

$$Q \;\succeq\; n_*\,\mathbf{1}.$$

At $P = 0.05, 0.2, 0.5, 0.8, 0.95$ the least eigenvalue equals n_* to machine precision and the transformation is canonical to 10^{-16}. Every member of the family has equal in- and out-region occupations, which is what the contact condition of §2.2, each mode half in each component where the sheets meet, requires on a symmetry-complete block, so such a state satisfies $\langle N_+ \rangle \geq n_*$. That equality reaches well past the family, and a short argument establishes it. Θ is antiunitary and exchanges the two regions, so $\Theta N_+ \Theta^{-1} = N_-$; for an antiunitary map $\mathrm{Tr}(\Theta A \Theta^{-1}) = \overline{\mathrm{Tr}\,A}$; and a Θ-invariant state has $\Theta\rho\Theta^{-1} = \rho$. Then

$$\langle N_- \rangle = \mathrm{Tr}(\rho N_-) = \mathrm{Tr}(\Theta\rho\Theta^{-1}\,\Theta N_+ \Theta^{-1}) = \mathrm{Tr}(\Theta\,\rho N_+\,\Theta^{-1}) = \overline{\langle N_+ \rangle} = \langle N_+ \rangle,$$

the last step because $\langle N_+ \rangle$ is real. So $\langle N_+ \rangle = \langle N_- \rangle = \langle Q \rangle$ for every Θ-invariant state, mixed, entangled or non-Gaussian, and the step from the operator bound to the observable needs nothing beyond invariance.

Integrating n_* with $P = e^{-x^2}$ returns the $I = 0.0127597$ of §2.2. The operator bound and the equality both hold on three hundred random mixed states and fail as soon as invariance is dropped.

The identity also has a limit, and the limit is informative. It works because Θ *exchanges* two things, so there is a second operator for the identity to equate. Where Θ *fixes* the object instead, the same identity constrains nothing: a squeezed vacuum with real r has real Fock amplitudes and is Θ-invariant at every r, while its occupation $\sinh^2 r$ runs freely, and only giving those amplitudes a phase makes the invariance bite. So the fold selects where it exchanges and not where it fixes, and the bang is the first case because the two branches are two things. Checked both ways on explicit states, at 10^{-16} against 0.155.

One consequence carries further than this section. A squeeze is measured by its occupation, $\langle N\rangle = \sinh^2 r$, so a selected occupation is a selected squeeze, at the crossing $r = \operatorname{arcsinh}\sqrt{n_*}$, running from 0.112 at $P = 0.05$ to 0.658 at $P = 1$. Since $n_* \le \frac{1}{2}$,

$$r \;\le\; \operatorname{arcsinh}\frac{1}{\sqrt{2}} = 0.6585,$$

a pure number with no parameter in it, and overshooting it would need $n_* > \frac{1}{2}$, which $(1-\sqrt{1-P})/2$ reaches only at $P = 1$. Reading that as a ceiling on the fold's two-sheet correlation at any later epoch needs two things this paper does not establish: that the crossing's invariance reaches the mode in question, and that its squeeze is inherited rather than regenerated. What does not depend on either is that the occupation the fold selects and the squeeze are one quantity, so the amplitude is not a free parameter.

At the other end of the fold, a horizon, the squeeze comes out determined, and the step that does it is §2.1's. Because the fold's map is the square root of the thermal transformation at the primitive period, a state in equilibrium at a bifurcate horizon has its cross-sheet correlator equal to the direct one shifted by half a period. A hole formed by collapse has no such horizon, so the statement covers the cosmological horizon and the two-sided class only. For one mode that shift gives $W(t - i\beta/2) = \cos\omega t/[2\omega\sinh(\beta\omega/2)]$, real and even, the two exponentials collapsing because $(1+n)e^{-\beta\omega/2}$ and $ne^{\beta\omega/2}$ are equal. The state carrying that correlator is the thermofield double, a two-mode squeezed vacuum with $\tanh r = e^{-\beta\omega/2}$ and $\sinh^2 r = 1/(e^{\beta\omega}-1)$ exactly. So the squeeze at a horizon is fixed by the mode frequency and the surface gravity, with nothing left to choose, subject to the fold being $J\circ P_\perp$, which leaves each sheet's own statistics alone and puts $(-1)^\ell$ on the cross term.

Equilibrium does not have to be assumed separately. A state can be built with no KMS property at a horizon, so nothing here shows that every state is thermal. What the Hadamard condition gives, and the fold has it already since §3.6 computes the image stress from the Hadamard parametrix, is that the state-dependent part of a two-point function is smooth while the singular part is fixed by the geometry alone. The temperature comes from the singular part. On a patch of proper size r the state-dependent part reaches the ratio $W_{\rm reg}/W_{\rm sing}$ only at order r^2, which holds on two unrelated families. A thermal state at temperature T gives $\pi^2 r^2 T^2/3$ with the exponent fitted at 2.000000 over five decades of r, and a classical background field gives $4\pi^2 r^2\phi_{\rm cl}^2$ with the same exponent. The balance is already the leading order in r, so a relative error of order $(rT)^2$ in the temperature sits one order below the field equation the balance returns, in the same way that the local fold of §3.6 is an isometry to second order and fails at third. Alter the singular coefficient instead, which is what a state outside the Hadamard class does, and the ratio stops vanishing, it goes to the alteration, at exponent zero.

The limit applies only for small r: a stellar-mass horizon sitting in the present microwave background has rT of order 3×10^6, so the expansion says nothing about a whole astrophysical horizon and everything about the shrinking patch the balance actually uses. What is assumed is the Hadamard condition, and equilibrium follows from it to the order used. The transverse parity is not an assumption of the same kind, because §2.1's map is forced: $J\circ P_\perp = -\mathrm{Id}$ exactly on the embedding, while J alone fixes a whole two-sphere and a rotation put in place of the parity fixes two points, so the $(-1)^\ell$ on the cross term is forced.

The two ends meet at one frequency. The crossing's ceiling $\operatorname{arcsinh}(1/\sqrt{2})$ is exactly the horizon's own value at $\beta\omega = \ln 3$, because $\sinh r = 1/\sqrt{2}$ gives $\cosh r = \sqrt{3/2}$ and hence $\tanh r = 1/\sqrt{3} = e^{-\ln 3/2}$. Modes softer than $T\ln 3$ sit above the ceiling and harder ones below it.

The inequality assumes no pure state, no Gaussian density matrix and no factorisation of the global state over momenta: a block reduced from an entangled multimode state is still a density operator and the inequality applies to it. It also survives the sum over modes, which is what the abundance needs. $\langle Q_k \rangle$ depends on the reduced state ρ_k alone, every reduced state is a density operator however the global state is correlated, so $\sum_k \langle Q_k \rangle \geq \sum_k n_*(p_k)$ with no factorisation assumed. Entanglement across momenta cannot lower the production integral, checked against generic entangled multimode states and against the state that saturates the bound exactly. What it does assume is free Bogoliubov evolution, the stated asymptotic particle definitions and equal in- and out-region number expectations; it is not a theorem about the interacting theory, and at $P = 1$ the even block is degenerate, so the zero-momentum limit is excluded from any uniqueness statement. Within those assumptions no admissible state returns a smaller production integral than the least-occupied member. Since $M_1 \propto I^{-2/5}$, every other state returns a smaller mass, and

$$M_1 \leq 491.6 \pm 2.0 \text{ PeV}, \qquad E_\nu = M_1/2 \leq 245.8 \pm 1.0 \text{ PeV},$$

with equality requiring $\eta(p) = 0$ wherever $\cos\lambda$ and the integration weight are nonzero. The least-occupied member is the case that saturates the bound, and no principle picks it out, and the selection criteria of Boyle, Finn and Turok and of Nadal-Gisbert, Navarro-Salas and Pla remain the additional input they were. Figure 2 shows the band, the floor under it, and how the abundance match reads the two.

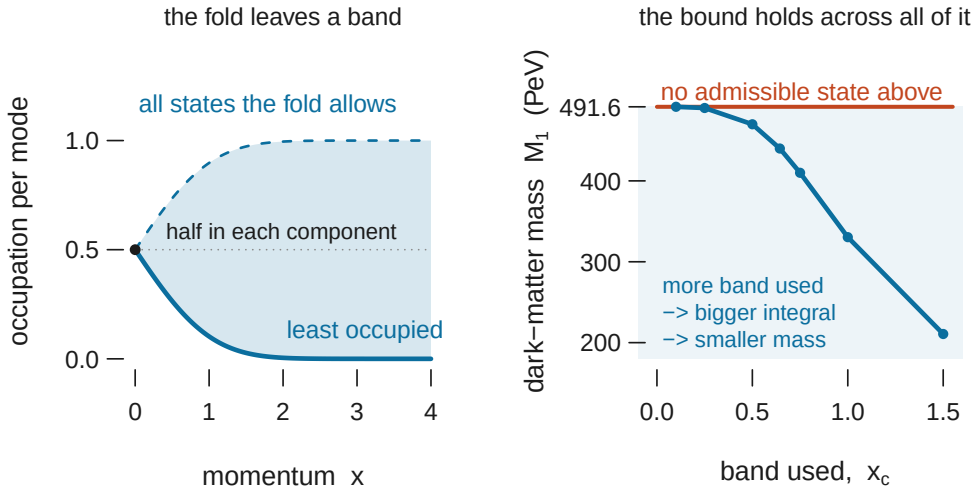


Figure 2. Why the mass is a ceiling. Left: Θ-invariance forces every mode half into each component at the bang, so the band of allowed occupations has zero width in the deep infrared and opens with momentum; the solid curve is the least-occupied member and the dashed one its particle-hole conjugate. Right: the mass each member of the family returns. Every excursion into the band raises the production integral, and M_1 falls as $I^{-2/5}$, so the least-occupied member sits at the top.

Unitarity makes the family symmetric about a half, $n_{\max} = 1 - n_{\min}$, checked to seven figures. At $p = 0$ the Hamiltonian is diagonal, no mixing occurs and $n = 1/2$ for every μ, so the band has zero width in the deep infrared. The upper branch is inadmissible on its own, since $n_{\max} \to 1$ gives a divergent number density. Table 1 quantifies the worst admissible excursion, a state at the band edge out to x_c and at the minimum beyond.

Table 1. The size of the surviving freedom.

x_c	$I/I_{\min}$	M_1 (PeV)	line (PeV)
0.10	1.000	491.6	245.8
0.25	1.008	490.1	245.0
0.50	1.119	470.0	235.0

x_c	$I/I_{\min}$	M_1 (PeV)	line (PeV)
0.75	1.574	410.0	205.0
1.00	2.700	330.4	165.2
1.50	8.312	210.7	105.4

Every excursion raises I and lowers the mass, and none can raise it, so $M_1 \leq 491.6$ PeV is the endpoint of the family the fold defines and not only of the family Boyle, Finn and Turok parametrise. The endpoint falls to KM3NeT's reconstructed 220 PeV median at $x_c = 0.644$. That sets a scale for x_c without limiting it. The median assumes an incident E^{-2} spectrum rather than a decay line, the event's origin is not established, and the band-edge profile is a worst case no real state need realise.

What regularity does and does not do. The Hadamard condition fixes the short-distance structure of the state. In the seed convention used here its leading large-momentum requirement is

$$\frac{c_+(0)}{c_-(0)} \sim \frac{i\gamma}{4p^2}, \qquad \gamma = ma_1,$$

with higher terms fixed by the adiabatic expansion. The bang-adiabatic member misses it and has an occupation tail proportional to p^{-4} whose energy integral diverges logarithmically, so it is excluded (Appendix B.1). Regularity does not select within what remains. For any smooth f that decays fast enough, replacing μ_* by $\mu_* + f$ preserves both the symmetry and every ultraviolet asymptotic coefficient while raising the occupation by $\sqrt{1-P}\sin^2(f/2)$; the freedom need not have compact support, only rapid decay, and the explicit CPT-invariant families of [22] exhibit it. That is why the ceiling has to be an operator bound and cannot be a selection. Among constant phases and the special values, only $\mu_*(p)$ cancels the boundary term down to the Gaussian (Appendix B.2).

The ceiling does not depend on the state at large momentum. The production integral runs to infinite momentum, and the state has to be specified there too. Half of I comes from below $x = 1.03$ and 99.9 per cent from below $x = 2.83$; truncating the state above $x = 3$ moves the endpoint by 0.076 PeV and above $x = 4$ by less than a thousandth of a PeV, against a quoted width of 2.0. The abundance is an infrared quantity. An ultraviolet condition acts on a different moment of the same occupation: the production integral is the $m = 2$ moment and converges even for a p^{-4} tail, since x^2x^{-4} is integrable, while the energy is the $m = 3$ moment and grows logarithmically for that tail, from 0.21 to 0.46 as the cutoff runs from 10 to 300, against 0.1358 for the adopted state. The ultraviolet decides which family is admissible; the infrared decides the number.

3.2 Neutrino line, the event, and the test that bites

The two-body line sits at $E_\nu \leq 245.8 \pm 1.0$ PeV. KM3NeT's reconstructed median of 220 PeV [21] lies 25.8 PeV below the endpoint, nearly twenty-six times the endpoint's width, so the comparison is not lost in the error bar. That width is the endpoint's, and the event's is far larger. KM3NeT quotes a 68 per cent interval of 110–790 PeV about the median, wide and strongly asymmetric in log energy, and roughly 46 per cent of that posterior lies above 245.8 PeV: 47 per cent on a split form matched to the upper half, 44 on the lower, and every intermediate choice between. That comparison inherits KM3NeT's assumption of an incident E^{-2} spectrum rather than a decay line, and it is an approximate posterior tail under assumed shapes, not a likelihood preference over an astrophysical population, which we do not calculate. The event neither confirms the endpoint nor refutes it: refutation needs an event confidently above the bound, and a posterior this wide cannot supply one. The median sits 10.5 per cent inside an allowed range, and a bound the data respects is a test that could have failed.

KM3NeT has since tested the decay reading itself. Their analysis of KM3-230213A as heavy dark-matter decay prefers a mass above roughly 100 PeV at 95 per cent confidence in every scenario they consider, with best-fit lifetimes of 10^{26} to 10^{27} s, and reports that those preferred regions sit in tension with bounds from other neutrino telescopes and from gamma-ray observations [31]. Their preferred scale is compatible with

the endpoint here, a weak agreement because the endpoint is a bound. What they report a tension with is the decay interpretation, which this paper does not adopt. Their channels are $\nu\bar{\nu}$, $\tau\bar{\tau}$ and $b\bar{b}$, and the mixed $h\nu/Z\nu/W\ell$ structure of §2.4 is not among them, so we neither claim their result transfers nor claim it does not. It is a live constraint on a reading adjacent to ours.

Borah, Das, Okada and Sarmah [23] reach the same kinematics from the other side. They select a heavy right-handed neutrino of 440 PeV so that its decay reproduces the observed 220 PeV, and report that the required lifetime saturates existing gamma-ray bounds. Their mass differs from M_1 by ten per cent. The decay interpretation is theirs and we claim no priority for it, nor do we suggest they hold it exclusively among heavy dark-matter readings of the event. What differs is the direction of inference: they choose the mass to fit the measured energy, while M_1 is fixed by the abundance and the half-mass energy follows.

What the test assumes. Shower energies at these scales are reconstructed to tens of per cent. At thirty per cent the detector width at the endpoint is 73 PeV against the model's 1.0, the test is limited by detector resolution. A fixed per-event threshold lets the sample-wide false alarm grow with the sample. Under the model's own endpoint the per-event crossing probability is 3.2×10^{-5}, and a conforming population trips such a rule 0.3 per cent of the time at a hundred events, 1.3 per cent at four hundred and 32 per cent at the twelve thousand the weakest case would need. Calibrated to a five per cent sample-wide false alarm, the operating characteristic is in Table 2.

Table 2. The calibrated endpoint test.

events	threshold (PeV)	power at $r = 2$	$r = 3$	$r = 5$
30	405	0.66	0.87	0.94
100	465	0.88	0.99	1.00
300	520	0.98	1.00	1.00
1000	580	1.00	1.00	1.00

A hundred reconstructed events at these energies separate this endpoint from one twice as high nine times in ten, at a five per cent chance of a false alarm. Neutrino telescopes have nothing like those numbers at these energies, so the population test cannot be run at present statistics.

The single-event test is the one that can fire now. For events drawn from a spectrum truncated at the endpoint and smeared by the response, the chance that a member of a conforming sample reconstructs three resolution widths above the endpoint is below one per cent at ten, twenty and thirty per cent resolution. One event clearly above the endpoint and securely assigned to the decay component refutes the implementation, and the weak link is the assignment.

What "securely assigned" requires. Flavour cannot do the assigning, since the matter rule leaves the Yukawa structure and with it the flavour ratio free. Direction can. Averaged over solid angle, the NFW column is 10.0 GeV cm^{-3} kpc in the hemisphere toward the Galactic Centre against 4.3 away from it, so decay places 69.9 per cent of its events in the near hemisphere against fifty for isotropy, and a binomial separation needs 58 events at three standard deviations and 159 at five. A ratio at exactly zero degrees is not quoted, since the NFW cusp makes it depend on the inner cutoff rather than on the halo. Those numbers sit at the same scale as the endpoint test, so one population of order 10^2 to 10^3 events supplies both the endpoint statistics and the directional assignment. Event counts assume an E^{-2} spectrum with a 50 PeV lower cutoff and a lognormal response, and the directional figure assumes pure Galactic decay with uniform exposure; neither carries backgrounds, extragalactic flux or detector acceptance, so they set the scale of what the falsifier demands and do not forecast when it will be met. One per-event discriminant exists and is weaker than it looks: an association with a transient by time and direction excludes the decay component, since a halo has no transients, but a steady, obscured or undetected source has no counterpart either, so non-association is necessary for the assignment and nowhere near sufficient.

The endpoint bounds the modelled two-body decay component and not the neutrino sky. Unrelated astrophysical events may exceed it at any energy, and testing it needs a specified lifetime, flavour and flux with propagation and detector response. What remains uncertain is the family, the radiation and entropy history, a stable decoupled sector and no later dilution.

The last of those has a number, and it is about as tight as the compact-object one above. Since $M_1 \propto (\rho_{\mathrm{DM},0}/s_0)^{2/5}$, a late release multiplying the comoving entropy by γ dilutes the produced ratio by γ, so matching the observed abundance needs a production γ times larger and the line rises as $\gamma^{2/5}$. The direction is opposite to a compact-object fraction, which lowers it. A release of 1.02 per cent moves the line by its own width, against the 1.01 per cent of the dark matter in compact objects that does the same, and a release by a factor of two would put the line at 324.3 PeV. The two heavier right-handed neutrinos are where such a release would come from, and the asymmetry they drive is ordinary thermal leptogenesis rather than anything the fold supplies, so their masses and lifetimes are not fixed here and neither is the size of any release. The line as quoted assumes none.

Nothing in the decay model reaches the ultra-high-energy cosmic rays, and the reason is kinematic. A decay product cannot exceed its parent, the whole dark-matter mass is 491.6 PeV, and Amaterasu was recorded at 244 ± 29 EeV [36], about five hundred times that. The same holds for every event at 10^{20} eV and above. Those arrivals are not this particle's and the fold says nothing about them. The ultra-high-energy neutrino is a different object at a different energy, and it is the one the model speaks to.

3.3 Neutrino-mass floor, and the exit the model has shut

Exact stabilisation leaves one light neutrino massless in the stated seesaw approximation. With normal ordering and $m_1 = 0$ the measured splittings fix

$$\Sigma m_\nu = \sqrt{\Delta m^2_{21}} + \sqrt{\Delta m^2_{31}} = 58.78 \pm 0.32 \text{ meV},$$

of which Δm^2_{31} supplies 89 per cent of the width; inverted ordering with $m_3 = 0$ gives 100.9 meV, and $m_{\beta\beta} = 1.5$–3.7 meV. The sum is obtained without fitting to any absolute-mass measurement, and it is a floor, it cannot be lowered without giving up the rank deficiency that produces it. Figure 3 places it against the cosmological bounds and the dark-energy constraint the same commitment carries.

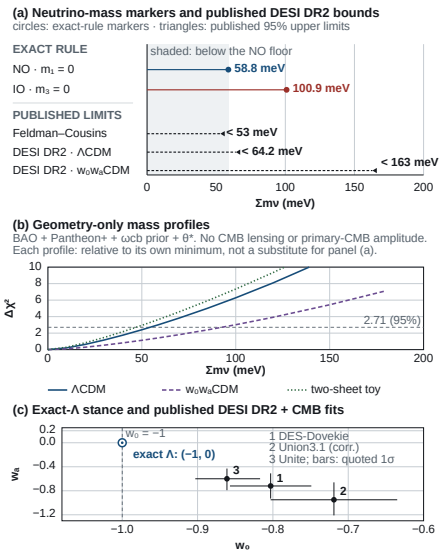


Figure 3. Neutrino mass and dark-energy context. *(a)* The massless-lightest-neutrino markers in the exact stabilising-rule sector at 58.8 meV (normal ordering, $m_1 = 0$) and 100.9 meV (inverted, $m_3 = 0$), against DESI DR2 BAO + CMB bounds from Elbers et al. [12]: 64.2 meV for ΛCDM, the cold-dark-matter cosmology with a constant Λ, and 163 meV for $w_0 w_a$CDM, the same with dark energy allowed to evolve, and the 53 meV Feldman-Cousins limit. Below 58.8 meV the shaded region is kinematically forbidden. *(b)* The three $\Delta\chi^2$ curves are a geometry-only reproduction using BAO, Pantheon+, an ω_{cb} prior and θ_*, with no CMB lensing or primary-CMB amplitude; they show the relative shape of the three models and are weaker than the published

bounds in panel (a). *(c)* The adopted exact-Λ commitment $(w_0, w_a) = (-1, 0)$ against three published fits incorporating DESI DR2 BAO and CMB with different supernova samples and quoted 1σ uncertainties: DES-Dovekie $(-0.803 \pm 0.054, -0.72 \pm 0.21)$ (arXiv:2511.07517), corrected Union3.1 $(-0.719 \pm 0.084, -0.95^{+0.29}_{-0.26})$ (arXiv:2601.19424), and Unite $(-0.861^{+0.044}_{-0.042}, -0.60^{+0.17}_{-0.19})$ (arXiv:2609.05053, which also includes DES BAO), and the $w_0 = -1$ phantom divide.

A floor on the neutrino mass constrains structure growth as well as cosmology. Free-streaming neutrinos suppress small-scale power by about $8f_\nu$, with $f_\nu = \Sigma m_\nu/(93.14h^2\Omega_m)$, which at the floor is $f_\nu = 0.0044$ and 1.76 per cent in S_8. None of that separates the fold from ΛCDM, because the same sum in ΛCDM does the same thing. The fold fixes the value of Σm_ν and does not change what a given value does. Against the baseline anyone quotes the difference is smaller still, since the standard S_8 is reported for a fit already assuming 60 meV, so the comparison is 58.8 against 60 and moves S_8 by 0.036 per cent. The fold therefore makes no S_8 prediction distinguishable from ΛCDM at the same neutrino mass, and it does not bear on the S_8 tension in either direction. Solving instead for the sum that would cancel a model excess measured against a massless baseline returns something near the floor, and that comparison is invalid whatever it returns, since the same neutrinos lower the baseline by the same amount.

Any bound on the sum depends on the dark-energy model assumed when deriving it. In Elbers et al.'s DESI DR2 analysis the same data give $\Sigma m_\nu < 64.2$ meV under ΛCDM and < 163 meV under w_0w_aCDM, because a relaxed expansion history can absorb the suppression a neutrino mass would produce. That is the standard way a model under neutrino pressure makes room, and this one cannot use it. For dark energy we adopt an exact cosmological constant, and for the restricted metric-only action class of Appendix D.2 the closure conditions give $p_\Lambda = -\rho_\Lambda$ and no exchange of energy with matter, $Q = 0$, even in a decelerating matter-filled background. The implementation is pinned by two separate commitments against one dataset: the floor cannot move down, and the branch where the bound relaxes by a factor of two and a half is closed. Panels (a) and (c) of Figure 3 are therefore not independent tests.

Under ΛCDM the window between the floor and the bound is 58.8 to 64.2 meV, a width of 5.4 meV or nine per cent of the floor. That bound is a posterior upper limit, with a tail above it. Three toy shapes from §2.5 put between 6.4 and 13 per cent of the posterior mass above 58.8 meV; truncating the flat toy at the limit instead of calibrating it at the ninety-fifth percentile would make 64.2 meV its hundredth and put the top of the range at 8.4 per cent. A spread this wide shows that one upper quantile does not pin a tail fraction. The profile likelihood is sharper, and used as §2.5 describes it puts approximately 6.8 per cent above the floor, near the bottom of the toy range. So the floor is allowed, 5.4 meV below the limit, and it occupies the part of the posterior the data least prefer. Elbers et al.'s own physical normal-ordering analysis quotes a considerably weaker bound than the degenerate-mass baseline, so these percentages are not the probability that the model survives.

The comparison a point prediction against a bounded measurement wants is a partition, and with the profile it is computable. With two propagated widths as the half-width, on the ground that a region of practical equivalence should hold values no measurement could separate from the floor, the region is [58.14, 59.41] meV. Table 3 partitions the profile posterior against it.

Table 3. Where the DESI DR2 profile posterior sits relative to the floor's region of practical equivalence.

region	posterior mass
below the equivalence region	0.929
within it	0.005
above it	0.066

Ninety-three per cent of the proxy posterior lies below a region the floor cannot be distinguished from. That remaining 7 per cent is not the model's share: exact stabilisation predicts the sum at the oscillation minimum and not every larger sum, so the 6.6 per cent above the region belongs to no prediction made here, and the 0.5 per cent inside it is not a survival probability. No decision rule is attached to the partition and none should be read into it. The half-width is a judgement, stated so it can be disagreed with; anyone preferring one or three sigma can recompute it from the two numbers above. What would settle the comparison is the

same partition of the survey's full posterior chain, which we do not hold, and the method is recorded here so that anyone holding the chain can apply it.

The Feldman-Cousins limit of 53 meV sits 5.8 meV below the floor, and on that construction the prediction is already excluded. We do not read it as an exclusion. A frequentist limit below the oscillation minimum is a statement about every normally ordered spectrum, since 58.8 meV is the kinematic floor for normal ordering whatever the cosmology, and such a construction is in conflict with the oscillation measurements whatever cosmology accompanies them. That tension is shared with oscillation data, not removed by it, and it is currently read as a feature of the data rather than of any one theory. We claim only that the conflict is not specific to this cosmology and that the bound is conditional on the analysis that produced it.

On the dark-energy side the same data press the other wall. Three published fits in Figure 3(c) sit 3.2 to 3.6 standard deviations from $w_0 = -1$ in the marginal w_0 direction. What is specific to this implementation is the absence of an exit.

A timescale follows. Elbers et al.'s $\Sigma m_\nu < 64.2$ meV sits 5.4 meV above the 58.8 meV floor, so the implementation is in tension once the published bound tightens by more than 8.4 per cent. On the crude scaling of a mass bound with the inverse square root of effective volume at fixed systematics, that is a volume growing by a factor of about 1.19. We offer no forecast: added data can move an upper limit in either direction, and the realised bound depends on the dataset combination, the priors and the systematic floor, none of which we model. What the arithmetic supplies is the size of the move that would settle it, and it is small. This pair is testable at the next DESI release on quantities already published, while the horizon tests of the companion wait on a ringdown measurement that does not yet exist at the required precision: GWTC-3's damping-time constraint [20] admits the companion's prediction of zero deviation under one reading of its posterior and not the other, and roughly twice the present exposure separates them. If the paper is wrong about the matter sector, that is where it will show first.

3.4 A closed dark sector

The relic abundance, once it fixes M_1, leaves no room in the dark matter: every gram is the sterile neutrino, and anything else massive competes for the same total. At 491.6 PeV each particle weighs 8.764×10^{-19} kg, so the measured dark-matter density is met by 2.6 particles per cubic kilometre at a mean spacing of 730 m, where a hundred-GeV WIMP at the same mass density would sit 4.3 m apart. This dark matter is not a fluid on any scale an instrument spans, and that is the picture behind the event count in Appendix D.1.

What presses on that closed sector is the population of compact red sources the James Webb Space Telescope (JWST) has found at $z \sim 4$ to 9, which appear to host black holes heavy for their epoch and which little of the literature is yet fitted to. Growth is not the difficulty. Eddington-limited accretion from $z = 20$ to $z = 7$ allows 11.6 Salpeter e-folds, so an $88\,M_\odot$ remnant reaches $10^7 M_\odot$ if it accretes at the limit throughout; seeding at $z = 10$ instead demands $3.1 \times 10^4 M_\odot$. The pressure in the literature is on sustaining that accretion, and heavier seeds relieve it. The relic itself does not help make seeds. Shot noise from a discrete particle gives a fractional seed overdensity $\sqrt{m/M}$, largest for a heavy particle and still only 2.1×10^{-27} at $10^5 M_\odot$. A seed needs of order 10^{-3}, so that route falls short by twenty-four orders of magnitude, and by twenty-seven against an overdensity of order one. The seeds must be astrophysical or primordial; what this closes is particle shot noise alone.

There is room for them, with margin. With one seed per host and a comoving host density of $10^{-4}\,\mathrm{Mpc}^{-3}$, seeds of $10^5 M_\odot$ carry a fraction 3×10^{-10} of the dark matter, and even generous variants stay below 10^{-5}. Since $\rho \sim I M_1^{5/2}$ gives $M_1 \propto (1-f)^{2/5}$, a fraction $f = 10^{-3}$ moves M_1 by 0.04%, well inside the ± 2.0 PeV of §2.3. A seed population ample enough to account for every such source perturbs the mass by far less than its own uncertainty, and the paper neither needs those sources nor is troubled by them.

What there is no room for is a dark sector made of black holes. At $f = 0.1$ the mass falls to 471.3 PeV and the two-body line to 235.7 PeV; at $f = 0.5$, to 372.6 and 186.3 PeV; at $f = 0.9$, to 195.7 and 97.9 PeV. The line is the observable of §3.2, and the direction runs the way one would not guess. A compact-object fraction lowers the line, and KM3NeT's reconstructed energy is below the $f = 0$ prediction, so such a fraction

moves the line *towards* the measurement. Solving $245.8(1-f)^{2/5} = 220$ gives $f = 0.242$: a quarter of the dark matter in compact objects would put the line exactly on the reconstructed energy. We record that as arithmetic only, and the constraints that bear on a compact fraction that large at these masses are somebody else's to apply.

A JWST result and a neutrino-telescope result are formally linked, since a compact-object fraction of the dark matter lowers the line as $(1-f)^{2/5}$, but the link is not reachable. To move the line by its own width of 1.0 PeV takes $f = 0.01014$, 1.01 per cent of the dark matter. At the seed density the little red dots imply, the line moves by 3×10^{-8} PeV, seven orders below its own width; the aggregate seed mass would have to be 3.4×10^7 times larger, some 3×10^3 seeds of $10^5 M_\odot$ per Mpc^3 against the 10^{-4} observed, and even a thousandth duty cycle leaves it four orders short. What the closed sector does give is a ceiling on a primordial dark seed component, $nM_{\rm seed} \le \rho_{\rm DM} = 3.3 \times 10^{10} M_\odot\,{\rm Mpc}^{-3}$, nine orders above the observed seeds. Baryonic seeds and envelopes are ordinary matter and do not count against it.

3.5 Decoherence time and dark-matter mass are one number

The fold relates two time orientations, and the production integral that fixes the mass also fixes when the two stop interfering at the radiation bang. In the adopted massive-particle state, for one produced pair with relative phase ϕ_k,

$$|D_k|^2 = 1 - 4n_k(1-n_k)\sin^2\phi_k, \qquad -\langle \log|D_k|\rangle_{\phi_k} = -\log(1-n_k)$$

for $n_k \le \frac{1}{2}$. In general the phase-averaged exponent is $-\log\max(n_k, 1-n_k)$, Jensen's formula for the mean of $\log|a_1 + e^{i\varphi}a_2|$ over a period, and the two forms agree wherever $n_k \le \frac{1}{2}$. Thus $\overline{\Gamma} = -\sum_k \log(1-n_k) = RN$ with N the produced-particle count, and in a Hubble volume

$$\overline{\Gamma}_H = \frac{4}{3\sqrt{\pi}}\, c_G\, R\, I \left(\frac{M_1}{H}\right)^{3/2}, \qquad t_{\rm dec} = \frac{1}{2M_1}\left(\frac{4}{3\sqrt{\pi}}\, c_G\, R\, I\right)^{-2/3},$$

where c_G is the mode-counting factor and the threshold $\overline{\Gamma} = 1$ is reached when a Hubble volume contains a particle count of order unity. With the adopted mass and $c_G = 1$ per (p,h) mode the crossing is at 1.417×10^{-32} s; counting a Majorana pair with $c_G = 1/2$ gives 2.249×10^{-32} s. That factor-of-order-one convention is not a precision claim. The unaveraged exponent oscillates rather than saturates, so this is a phase-averaged comparison and not a proof of irreversible branch selection.

Section 2.3 gives $M_1 \propto I^{-2/5}$, so $I \propto M_1^{-5/2}$, and the crossing time is $t_{\rm dec} \propto M_1^{-1} I^{-2/3}$. Elimination of I between them leaves

$$t_{\rm dec} \propto M_1^{2/3},$$

a power with no free parameter. Given the two laws it is exact, eliminating I leaves $-1 + \frac{5}{3} = \frac{2}{3}$. Composing the two laws numerically over four decades in M_1 returns the same exponent to better than 10^{-15}, which checks that they are implemented as stated and not that the power is right. So one production integral at the bang sets both the particle mass and the time at which the two time directions stop interfering. Normalisation is a separate matter. R is a second functional of the occupation and moves with the state freedom of §3.1, giving $t_{\rm dec} \propto M_1^{2/3} R^{-2/3}$.

Table 1 shows how large that dependence is. At its $x_c = 0.644$ row, the one that reproduces KM3NeT's reconstructed energy, R falls by 24 per cent, the prefactor rises by 20, and the net decoherence time rises by 12. Table 1 runs further, and at its far edge, $x_c = 1.50$, R falls by 88 per cent, the prefactor rises by a factor of 4.1, and the net time is up by 133 per cent. Its state dependence is an order of magnitude larger than the row we quote, and we quote that row because it is the one the measurement picks out.

That is the variation along the band as the state moves off the minimum. At the endpoint there is none: the band has floor $n_{\min}$, so the minimum profile is the unique admissible state with $I = I_0$, which fixes R there. At a fixed I above the endpoint different admissible profiles give different R. Profiles built inside the band as $n = n_{\min} + \Delta w(x)$ with $\Delta = n_{\max} - n_{\min}$ and $w \in [0,1]$ are admissible at every momentum by construction, and two of them matched to the same I differ in the prefactor by about 9 per cent at $I/I_0 = 1.15$ and 15 at 1.30. Those examples show the freedom exists; they do not bound it, and no finite scan can. A measurement of either quantity constrains the other only once R and the counting convention are fixed. The relation inherits every assumption of the two sections it joins: the production state and radiation history of [10,11], the small-Weyl-coupling branch, the phase averaging and c_G, which moves the prefactor but not the power. Only the power is independent of those choices.

The closed-de Sitter throat is a separate clock, worked in the companion. A real no-boundary wavefunction contains expanding and contracting branches, and the companion obtains the reality from Θ-invariance rather than only from the no-boundary proposal. With one minimally coupled scalar on closed de Sitter the overlap of the branch states reaches e^{-1} when the scale factor is about twice its throat value, after a time of order H^{-1}, which defines a decoherence threshold, with no discontinuous birth of time. Appendix A.3's gauged spacetime inversion does not delete the odd-parity oscillators from the mode sum, since the branch states are Gaussian in each mode coordinate and so already invariant. The exponent carries the multiplicity N_f of identical minimally coupled scalar environments linearly, with threshold $aH = 1.954$ for one such scalar and $aH = 1.093$ at a multiplicity of 106.75. That multiplier is a scalar toy and not a Standard Model result, since different spins and couplings give different overlaps, and the $N_f^{-1/3}$ law is a large-A approximation. Direct summation gives $N_f(A-1)$ flat at about 14.8 across $N_f = 10^4$ and 10^5, so near $A = 1$ the threshold approaches as N_f^{-1}. A denser environment separates the branches sooner. Those two calculations use different backgrounds and different physical clocks.

What the fold leaves joined is a crossover. The produced pair is pure, so the entanglement entropy between its two members, a particle at p and an antiparticle at $-p$ in the out region, is $S(n) = -n\ln n - (1-n)\ln(1-n)$ and the mutual information is $2S$. At $x \to 0$ the Hamiltonian is diagonal, $n = 1/2$ exactly and $S = \ln 2$, its largest value. At large x the occupation is Gaussian and S collapses, 3.5×10^{-4} at $x = 3$ and 9.5×10^{-11} at $x = 5$. Half the maximum is reached at $x = 0.968$ and the mode-weighted total $\int x^2 S\, dx = 0.4509$ converges. Production at the bang makes maximally entangled pairs at long wavelength and unentangled ones at short, so what the fold leaves behind is a crossover with a scale rather than a surface with a location. Its scale is not a new input: the transition probability e^{-x^2} falls to $1/e$ at $x = 1$, within three per cent of the half-entanglement point, so the sheets stop being one object at the scale at which the crossing stops being abrupt. The two members of a pair are commuting subalgebras and tracing one out is legitimate; the two sheets at the bang are not, since $\eta < 0$ and $\eta > 0$ are one free field at two times, and no entropy between them is calculated. At a horizon the two-subsystem reading is available because J relates a wedge algebra to its commutant; at the bang it is not, and the sheet statement that holds there is the symmetry of §3.1.

3.6 Einstein's equations from the fold, with nothing quantised

Two of the fold's results have been carried separately and are one result. The fold's parity is the classical-quantum split, Φ_c being the fold-even half and Φ_q the fold-odd one. And at a bifurcate Killing horizon, one whose generator vanishes on a surface, the fold's map is the half-period shift, so the cross-sheet correlator is the direct one at $t - i\beta/2$. Those two facts together fix the weight of the quantum half and give it a value.

For a single mode, $\langle\Phi_c^2\rangle = \frac{1}{2}[W(0) + W_{\rm cross}]$ and $\langle\Phi_q^2\rangle = 2[W(0) - W_{\rm cross}]$, with $W(0) = \coth(\beta\omega/2)/2\omega$ and the half-period shift giving $W_{\rm cross} = 1/2\omega\sinh(\beta\omega/2)$ from the KMS condition alone. The ratio collapses,

$$\frac{\langle\Phi_q^2\rangle}{\langle\Phi_c^2\rangle} = 4\tanh^2\frac{\beta\omega}{4}.$$

So the weight of the quantum half is set by a temperature. Below the horizon temperature the ratio falls as $(\beta\omega)^2/4$ and the quantum half switches off, which is the classical limit reached in temperature rather than

in $\hbar$; above it the ratio saturates at 4. The two halves carry equal weight where $\tanh(\beta\omega/4) = 1/2$, that is at

$$\beta\omega = 4\,\mathrm{artanh}\,\tfrac{1}{2} = 2\ln 3,$$

exactly twice the frequency at which §3.1's crossing ceiling $\mathrm{arcsinh}(1/\sqrt{2})$ meets the horizon squeeze, and at a squeeze $\tanh r = 1/3$ which is the square of the $1/\sqrt{3}$ there. Both are the same statement about occupancy. Moving the fold's shift off a half period moves the equal-weight point off $2\ln 3$, so the number belongs to the fold and not to thermality alone.

Everything above treats the metric as a fixed background. That is usually a limitation. Here it is what is being shown: the field equations governing that background can be obtained from the same fold without the metric ever becoming an operator.

Jacobson's argument. Jacobson derives the Einstein equation from the Clausius relation, $\delta Q = T\,dS$, applied to local Rindler horizons, the horizons an accelerated observer sees [33]. Because that step carries the whole of what follows, it is re-derived here. It rests on four things: the Clausius relation itself, a temperature at every such horizon, an entropy proportional to that horizon's area with a universal coefficient η, and a horizon of the kind at every point in every null direction. The first is the thermodynamic postulate the argument is made of. Jacobson took two of the other three from quantum field theory, the temperature from the Unruh effect [37] and the area law from vacuum entanglement with a single cutoff, after Bombelli and coauthors and Srednicki [38,39], and the third, the horizons, from local flatness. Here each is re-derived from the fold's map. At a flat wedge that map is the CPT operator. Bisognano and Wichmann's modular conjugation of a wedge is CPT composed with a rotation by π in the transverse plane [28], and $P_\perp$ is that rotation, so $\alpha = J \circ P_\perp$ is CPT, with the usual $(-1)^F$ on fermions; on coordinates it is the $-$Id of §2.1. The derivation below is therefore Jacobson's, run on the map that also defines the folded cosmology, and the end of this section says what the fold adds to it.

Raychaudhuri's equation tracks how a bundle of null rays converges. Integrating it near the bifurcation surface, where the expansion and shear are first order so their squares are second, gives $\theta = -\lambda R_{kk}$ to within 4×10^{-4} at $\lambda = 0.05$ and visibly worse further out, as a first-order expansion should. The heat flux carries the same $\int \lambda\, d\lambda\, dA$, and the surface gravity cancels between the two sides, which is what lets a statement about one accelerated observer become a field equation. What is left is $2\pi T_{kk} = \eta R_{kk}$ for every null k.

That a symmetric tensor annihilating every null vector must be a multiple of the metric is linear algebra, and the check is: the constraint map built from sixty random null vectors has a one-dimensional kernel, its ninth and tenth singular values differing by 3×10^{15}, and that kernel is g_{ab} to 9×10^{-16}. Timelike vectors leave no kernel at all, so nullity is doing the work. The contracted Bianchi identity and $\nabla^a T_{ab} = 0$ then fix the remaining function, and the balance is $R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} = (2\pi/\eta)T_{ab}$ with Λ an integration constant of the derivation rather than a prediction of it.

Two objections. Two objections to the 1995 argument have to be met here. Eling, Guedens and Jacobson showed that the equilibrium Clausius relation fails once the entropy is not proportional to area with a universal coefficient, and that an entropy-production term is needed in its place [34]; their worked case is $f(R)$. In the fold, as in Jacobson's argument with its single cutoff, η is not available to choose. It comes out of the transverse mode count with the surface gravity cancelling, which the entropy paragraphs below compute, so the derivation is shown to sit in the case the equilibrium relation was written for.

Chirco and Liberati showed that the shear supplies an internal production term of its own, which they identify with tidal heating [35]. That term is second order at the bifurcation surface, where the expansion and the shear are both first order, and the Raychaudhuri measurement above says how far out that survives, the linear behaviour holds to 4×10^{-4} at $\lambda = 0.05$ and visibly worse beyond. What is still assumed is the equilibrium reading itself, taken near the bifurcation surface of each local wedge, and that is the first of the four things listed above, the one the fold does not supply. The fold, like the vacuum Jacobson used, leaves no freedom in η or in the temperature and leaves that assumption standing.

The temperature. The temperature comes first, and nothing in it is left to choose. §2.1's map is an involution and its action at a bifurcate Killing horizon is the half-period shift, so the equilibrium cross-sheet correlator

is the direct one displaced by $i\beta/2$, which is a thermofield double at $\tanh r = e^{-\beta\omega/2}$. No temperature is put in by hand, and none is in the Unruh effect either [37]: by Bisognano-Wichmann the two are one statement. In modular language the half-period shift is the $\Delta^{1/2}$ of Tomita-Takesaki theory, which makes the map the square root of the thermal transformation and lets primitivity pick the fundamental period.

What the step establishes turns on a distinction the companion draws and this section needs too. Squaring the map gives $\alpha^2 = 1$, which reads $W(t - i\beta) = W(t)$, the KMS condition, and sweeping the shift period the companion finds it fails by $O(25)$ at every value tried and holds only at $\beta = 2\pi/\kappa$ and its multiples, the minimised residual returning 6.2831853072 against $2\pi/H = 6.2831853072$. That match is a consistency check and not a second derivation of the temperature, since a csch^2 kernel carries period $2\pi i/H$ however it was obtained. What it does establish is the half stated above, that the fold's map is the square root of the thermal transformation.

That the condition admits no near misses is a separate point, and it is the one Jacobson's argument uses. Written as detailed balance the condition is $\widetilde{W}(-\omega) = e^{-\beta\omega}\widetilde{W}(\omega)$, and an occupation departing from the Bose factor violates it at first order in the departure, so no window of nearly-thermal states passes. The horizon temperature the Clausius relation is handed is therefore exact or absent, with nothing in between for an approximate state to occupy.

The horizons. The horizons come next, and all of them at once instead of a preferred family. In any orthonormal frame the reflection of a Rindler wedge is $\mathrm{diag}(-1,-1,+1,+1)$ and the transverse antipode is $\mathrm{diag}(+1,+1,-1,-1)$. Neither factor is frame-independent and their composition $-\mathrm{Id}$ is, so a single map is the fold of every Rindler wedge through its fixed point, at every boost and every orientation. Composing the two in two hundred randomly boosted and rotated frames returns $-\mathrm{Id}$ to 3×10^{-14}, while the reflections themselves move by order unity.

The surfaces Raychaudhuri acts on are the fold's own. A conjugate locus is a surface where a whole family of rays leaving one point refocuses. In the companion, the region where a point can reach its own fold image is bounded by one, at every charge and in every spacetime dimension. The fold's own contact surfaces are therefore null-focusing surfaces, the class of object Raychaudhuri and so Clausius act on.

The entropy. The entropy law decides whether any of this is circular. If $S \propto A$ were taken from Bekenstein-Hawking it would have come out of general relativity and been fed back into a derivation of it. Jacobson did not take it from there either: he motivated it from the entanglement of vacuum fluctuations across the horizon, as Bombelli and coauthors and Srednicki had [38,39], and that entanglement is what is computed here. Tracing out one wedge of the thermofield double the fold forces leaves an occupation $\sinh^2 r$, and with $\tanh r = e^{-\beta\omega/2}$ that is exactly the Bose factor, so the entanglement entropy of a wedge is the thermal entropy of its modes, $s = (1+n)\ln(1+n) - n\ln n$. A horizon's modes carry a transverse momentum, the transverse directions are translation invariant, and the mode count of a patch is therefore proportional to its area, which factors straight out and is flat-space field theory with nothing gravitational in it. The dependency graph is in §4.1, and beside it the four places this derivation could have failed.

Proportionality is half of what Jacobson asks for. He needs the same η at every horizon, and $\beta = 2\pi/\kappa$ differs from one to the next, so the occupation visibly carries the surface gravity. It cancels. In the observer's own proper frequency the Wentzel-Kramers-Brillouin (WKB) mode count below ω is $(1/\pi)\int_\epsilon^{\omega/\kappa k} \sqrt{\omega^2/\kappa^2\rho^2 - k^2}\, d\rho$, which carries κ in three places, and the Jacobian $d\omega = \kappa\, d\Omega$ cancels the $1/\kappa$ the density carries. Calculated with κ kept throughout, S/A does not move to one part in 10^9 while β runs from 251 to 0.025, four decades. This follows from the theorem that gives the fold its map. The wedge metric in boost coordinates contains no κ, and Bisognano-Wichmann makes the modular temperature 2π in boost time at every wedge, so $\beta\omega = 2\pi\Omega$ identically. Let the modular temperature depend on κ instead and S/A spreads by a factor of 21.6.

What S/A does depend on is the cutoff, as $1/\epsilon^2$ to four figures, and that joins two statements this section had been making separately: the coefficient is universal and it is divergent, of mass dimension two, which is exactly the object no structural input of dimension zero could return.

The cutoff dependence also attaches one condition, found by trying to break the result. Because of that $1/\epsilon^2$, a cutoff allowed to track the surface gravity as $1/\kappa$ moves the answer by a factor of 52 over the same

range, so η is the same at every horizon provided the ultraviolet cutoff is one length and not one per horizon, which is what a cutoff is. The mode count is not a condition: weighting the transverse density by $e^{-k\epsilon}$ or by $1/(1+k^2\epsilon^2)$ moves S/A from 2.137 to 1.629 and 1.931 and leaves the κ independence at four parts in 10^{10}, so the cancellation belongs to the boost structure and not to one prescription. So η is the same at every horizon given one cutoff, which is what Jacobson's single fundamental length already assumed. Whether the cutoff is finite and the same for every species is the input his later work names as still open [40], and it stays open here. What η is remains measured.

That last sentence separates this from induced gravity [44,43], which uses the same object. Induced gravity computes $1/G$ from a given matter content and cutoff, and stands or falls on whether the species count comes out; walked as a route to η here, it does not, and no such claim is made here. The argument needs only that the coefficient is the same at every horizon, which is the previous paragraph, and takes its value from experiment, which is what $8\pi G = 2\pi/\eta$ then reads as. Jacobson's step requires only the weaker claim, and only the weaker claim is made.

Because η is fixed by the modular temperature and a transverse mode count, and neither is cosmological, it carries no dependence on epoch or location, so nothing in the fold can make Newton's constant run, any more than a single fundamental length could in Jacobson's argument. That is a commitment of the same kind as $w_0 = -1$, arrived at the same way, and a securely measured variation in G would end this reading of the fold with nothing available to absorb it.

The sign of Newton's constant comes with it. Since $8\pi G = 2\pi/\eta$ and η is an entropy per unit area, which is positive at every frequency, $G > 0$, so gravity attracts here because entanglement entropy is positive. A priori that sign is free; Jacobson's $G = 1/4\hbar\eta$ fixes it the same way.

A horizon at every point. That leaves the fourth input. Jacobson needs a horizon at every point in every null direction, and took it from local flatness. The fold's own fixed points are isolated, so at a generic point there is no global fold to act. What the balance needs there is weaker than a global symmetry. It needs a map whose differential is $-\mathrm{Id}$, and it needs that map to be an isometry to the order the balance is computed at.

The geometry supplies both, everywhere. The geodesic symmetry $\exp_p v \mapsto \exp_p(-v)$ has differential $-\mathrm{Id}$ by construction, so it is the fold's own local model at any point of any spacetime. In normal coordinates the metric's quadratic term is built from R and its cubic term from ∇R, which makes that symmetry exact through second order with a third-order failure. Cartan's theorem says the failure is real wherever $\nabla R \neq 0$, so what has to be settled is how heavy it is.

Measured on a two-dimensional surface with Gaussian curvature K and $\nabla K \neq 0$, the residual $G_{\mu\nu}(v) - G_{\mu\nu}(-v)$ has exponent 2.9994 against 3, and a coefficient agreeing to five figures with the closed form $\frac{1}{3}(\nabla K \cdot v)(|v|^2\delta_{\mu\nu} - v_\mu v_\nu)$. On a surface whose cubic part is harmonic, so that the curvature varies while ∇K vanishes at the point, the cubic disappears and the exponent moves to 5.04, what the residual tracks is ∇R at the point and nothing else. Set against the second-order term the balance uses, the ratio is 1.4435 times the patch size and vanishes with it. The field equation is that limit, the same one Jacobson took by working in a locally flat patch.

The parity gives a second reason the balance cannot feel the failure. A map with differential $-\mathrm{Id}$ multiplies a rank-n tensor by $(-1)^n$, and the balance contains R_{kk}, T_{kk} and a transverse area element, every one of them even. The boost field is rank one and flips. That flip is the wedge swap, the modular generator is $H_R - H_L$, and $\beta = 2\pi/|\kappa|$ does not move. An odd-rank quantity would break, and the balance holds none.

The fold's own term. Running the balance on the fold's own horizons puts one further term into it, and whether that term is local decides whether the result is general relativity or something more. The fold's image contribution to the stress enters the flux alongside ordinary matter, so

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(T_{ab}^{\mathrm{matter}} + T_{ab}^{\mathrm{img}}[g, \Theta] \right).$$

If the second source were a local curvature polynomial it would renormalise G and nothing else, and this would be general relativity with a shifted coupling. It is not local. The image stress depends on the world

function between a point and its fold image, half the squared geodesic interval between the two, which is a function of two points and not of one, and the companion's exact case makes the failure explicit. On the Einstein static universe every local invariant is a constant, because the space is homogeneous, while $\rho_{\rm img} = (6\xi-1)/16\pi^2a^2(1+\cos\eta)$ runs with the separation η to the image. The best local fit leaves a residual of 139 per cent of itself, and the same fitter reproduces a genuinely local stress exactly.

That term carries no free parameter. Away from a caustic its size is fixed by the matter's departure from conformal invariance and by nothing else. It is proportional to $1-6\xi$ and vanishes for conformally invariant matter, and the same slot in the Hadamard coefficient carries m^2 for massive fields. That second slot is calculated here: at conformal coupling with a mass the null-null component is negative at every mass and close to proportional to m^2, so the sign the companion's A.15 needs holds through both slots rather than through the coupling alone.

The two slots also diverge at different rates as the caustic is approached, and the difference has a reason. In the distance to the caustic the coupling slot goes as $\delta^{-4.010}$, which is what the caustic-order rule stated below and two derivatives predict, and the mass slot as $\delta^{-2.037}$. It is one power softer because a mass carries one power of the world function, and one more because the conformal coupling is, on that geometry, exactly the value at which the leading mass parts cancel.

Outside every horizon the image term is suppressed to nothing that could be measured, by three separate mechanisms, since no theorem covers it. The companion's silence theorem says the cross-sheet commutator vanishes there, and the companion says in its own words that the theorem does not extend to the stress tensor. What does the work is that a point and its image are spacelike separated there.

A massive field's image correlator then carries e^{-md}, with md above 10^{16} for an electron at a stellar horizon and 10^9 for the lightest neutrino anyone proposes. A field light enough to escape that is light enough for the m^2 in the Hadamard slot to finish it, the worst case across the whole range sitting at 3×10^{-34} of the dark energy. And a massless conformal field has an empty slot and a Weyl-suppressed remainder falling as r^{-5}, which is integrable and leaves no deficit angle. Inside a horizon the term lives on the inner half of the interior and nowhere else. At the boundary of that region, the contact sphere at $r = r_h/2$, the companion finds it diverges as $\delta^{-3/2}$, and inside, where the interval is no longer spacelike, it continues to $i|\sigma|^{-3/2}$ and turns imaginary. An imaginary energy is a decay rate, and a decaying vacuum makes pairs, so the region where the two sheets touch makes matter and antimatter out of the causal structure with nothing added. That is the companion's result and not used here.

The sign there follows through, because for a null k the trace term and Λ drop out of the field equations, so $R_{ab}k^ak^b = 8\pi G\,T_{ab}k^ak^b$ with no residue and the two signs are locked together. A negative $R_{ab}k^ak^b$ is a failure of the null convergence condition, which is the hypothesis Penrose's argument uses to turn a trapped surface into an incomplete geodesic. So the one region where general relativity predicts its own breakdown is the one region where the fold departs from it, and which way that departure runs is settled below.

§4.2 gives the answer: the fold's term defocuses the congruence along which the two sheets would touch and focuses the radial one Penrose's argument follows, so the null convergence condition holds where he uses it and fails only along the contact direction. How far the departure reaches is a definite number too. An ingoing radial null congruence in the Schwarzschild interior is shear-free, focuses on its own, and saturates Penrose's bound exactly, reaching infinite convergence at affine parameter $2/|\theta_0|$ with no stress at all; it enters the contact region with $\theta = -4/r_h$, and the level the fold's term has to reach to bring that expansion back to zero inside the region is $11.5/r_h^2$, from $\operatorname{artanh}(1/u) = u$ at $u = 1.1997$.

How thick the shell is. The fold's term is not constant across the region, and it diverges on the contact sphere, which is the region's outer edge and therefore where an ingoing congruence enters. What takes a number is a thickness, the shell in which the fold's term is the larger of the two. Dimensions leave one combination of the two lengths available, $T_{ab}k^ak^b = \kappa m^2 D^{-p} r_h^{p-2}$, and the shell is $D_*/r_h = \left(8\pi\kappa(m/m_P)^2/11.5138\right)^{1/p}$, proportional to r_h and to nothing else.

The power decides everything about that thickness and κ almost nothing, since κ enters as its p-th root. That power is 5/2, and it follows from three things each measured on the one geometry where every quantity is a closed-form mode sum. The image Green function at a caustic of order n in D dimensions goes as

$\delta^{-(D-2+n)/2}$ in the distance to it, which is the caustic-order rule, so a stress built from two derivatives goes as $\delta^{-(D+2+n)/2}$. That predicts δ^{-4} there and the computation gives $\delta^{-3.95}$, which tests the rule on a stress.

The mass-dependent part of every quantity comes out one power softer than its massless counterpart, at 0.956, 0.976 and 0.968 of a power for the correlator and its two second derivatives, and the reason is the proper-time representation. A mass is a phase in the proper time, so to first order the mass part is the same integral with one extra power of s, and one extra power of s raises the world-function exponent by exactly one, checked against quadrature at six pairs (D, n) and against the exact massive integral as $m \to 0$.

What differs between a caustic and open space is how the world function vanishes. At a caustic of this kind it vanishes linearly in the distance, $\pi a\delta$ on that geometry and proportional to $M - r$ at a hole, so one power of the world function is one power of the distance. In flat space it vanishes as r^2 instead, so the same statement reads there as two powers and a log, which is the familiar flat-space answer.

The further power the mass slot appeared to lose belongs to the coupling. Where the Ricci tensor vanishes the null stress is $(1-4\xi)P-Q$ and the leading mass parts cancel only when $1-4\xi$ equals dQ/dP at the caustic. On a conformally flat geometry that ratio is $1/3$ and the cancelling coupling is therefore the conformal one, which is why the mass slot on that geometry sits two powers soft; at $\xi = 0$, $1/12$, 0.2 and 0.25 it keeps its own power of 3, and $\xi = 1/6 + 0.02$ already gives 3.01. So the caustic at a hole being order one, the leading stress is $\delta^{-7/2}$ and the mass part, which is the part that acts where the Ricci tensor vanishes, is $\delta^{-5/2}$.

At $p = 5/2$ the thickness is a definite length far above the Planck scale, and κ is no longer an unknown either. Figure 4 draws both halves of that, where the term acts, and how thick the shell is against the field's mass. Every ingredient of it is already here.

The caustic amplitude is $\Delta^{1/2} \to 3.9004\, M\, s^{-1/2}$, and it is settled: moving the Jacobi profile at fixed length and family volume on a non-symmetric surface of revolution leaves the heat kernel where it was, which leaves the length. That length is the arc of the geodesic's projection onto the sphere the caustic lives in, $\int r\, d\varphi = M(\pi + 2)$ along $r = M(1 + \sin\varphi)$, so $c_1 = \sqrt{\pi}\, M(\pi + 2)$. Three more go in: the two proper-time integrals, with the mass as one extra power of s; the world function's linear vanishing, with the coefficient $\tau^2/(M - r) \to 34.85$; and a contraction factor that is exactly one on the ingoing radial congruence, because $dr/d\lambda = -1$ there.

Assembled, $\kappa = 0.0039$, which shrinks the shell to 0.109 of its width at $\kappa = 1$, since the thickness goes as $\kappa^{2/5}$. At a solar-mass hole that gives 2.1 microns for the 491.6 PeV fermion of §3.1, 1.5×10^{-11} m for a top quark, 2.3×10^{-13} m for a proton and 5.5×10^{-16} m for an electron.

The order-unity factor that used to stand beside κ, from keeping only the leading derivative pairing, is measured at one. With $G = F(\sigma)$ the null-null second derivative is $F''(k \cdot \nabla\sigma)^2 + F' k^a k^b \nabla_a \nabla_b \sigma$, and the second piece carries two fewer powers of σ, so it cannot reach the leading coefficient. Setting the leading pairing against the companion's exact tower, in the geometry where its amplitude rule leaves nothing to choose, the two agree to twelve digits for the massless part and to 0.2 per cent for the mass part at the closest offset reached. So 2.1 microns is the number, with no decade of slack around it; reaching 5.3 would need κ inflated tenfold.

So the region where the fold parts company with general relativity is, for ordinary matter, a shell of nuclear thickness around the contact sphere, and for the fold's own dark-matter fermion a shell you could see. It widens as the hole does, since D_* is proportional to r_h, which puts the effect at its largest in the largest holes.

What it does not do is thin away to nothing as the mass falls, and saying why corrects the $1 - 6\xi$ law two paragraphs above. That law is exact where the parametrix holds and on the conformally flat geometry it was calibrated on, and it fails at a null caustic in a spacetime that is not conformally flat, which is what a contact sphere is.

There the coupling cancels outright: with $k \cdot k = 0$ the metric terms of the point-split stress drop, the leading behaviour of the image term is $C\sigma^{-3/2}$ so both second derivatives are dominated by the $\nabla\sigma$ pairing with $\nabla_{b'}\sigma$ minus the transport of $\nabla_b\sigma$, and $(1 - 2\xi)(-X) - 2\xi X = -X$ at every ξ. The mass and the curvature coupling both multiply the image term itself, two powers softer. Summing the tower at five couplings and

four masses returns one coefficient, -0.0592, with a spread of 1×10^{-5} across couplings and no zero at $\xi = 1/6$.

So a massless conformal field, for which the $1 - 6\xi$ law predicts nothing at all, has a shell of its own, and it is the floor under every other. Two derivatives on the companion's amplitude give $T_{kk} = -3.19 \times 10^{-3} M^{-1/2}(M-r)^{-7/2}$, and setting that against the same $11.5138/r_h^2$ leaves $D_{\text{floor}} = (9.85 \times 10^{-3})^{2/7}\, \ell_P^{4/7} r_h^{3/7}$, a geometric mean of the Planck length and the horizon radius carrying no matter content at all. At a solar mass it is 1.1×10^{-19} m and at a billion solar masses 7.8×10^{-16} m, sixteen orders above the Planck length. It moves nothing above, because two extra powers of divergence arrive with a coefficient the mass beats by 3×10^{46} at the fermion's own shell and by 10^{13} at the electron's. What it changes is the shape of the claim. The shell has a floor, and nothing the fold can be made of falls through it.

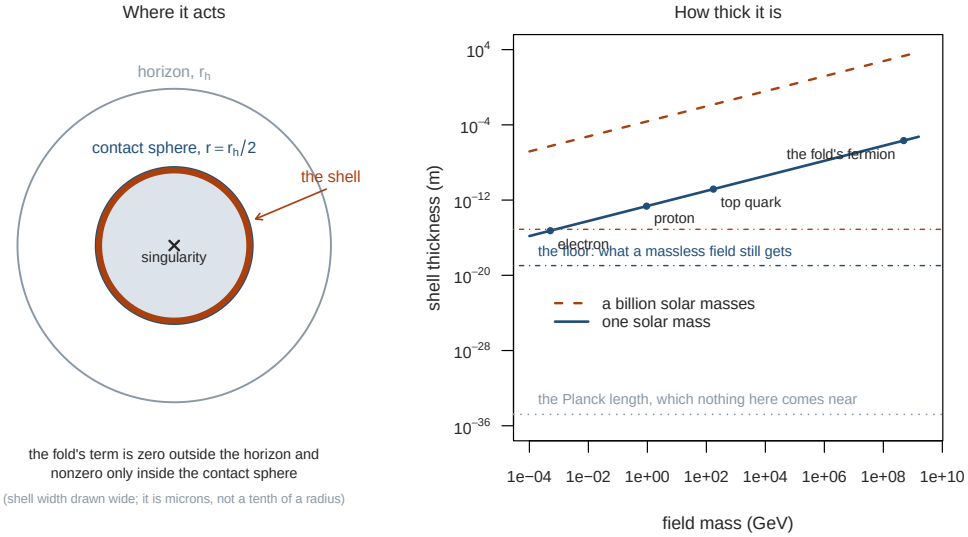


Figure 4. Where the fold leaves general relativity, and how thick that place is. Left: the fold's extra term is suppressed to nothing measurable outside the horizon and acts only inside the contact sphere at $r = r_h/2$, and the shell drawn in at the edge of that region is where it is the larger of the two terms acting on an infalling congruence. The shell is drawn wide to be visible; its true width is microns. Right: that thickness against the field's mass, at one solar mass and at a billion, with the Planck length marked. Nothing on the plot comes near it, which is the result: the departure is a real length for every field there is. The two dashed horizontals are the floor the mass-independent part of the same stress leaves, 1.1×10^{-19} m and 7.8×10^{-16} m, which the sloping curves meet only for a field lighter than anything in the Standard Model and below which nothing goes. Both curves are $D_*/r_h = (8\pi\kappa(m/m_P)^2/11.5138)^{2/5}$ with $\kappa = 0.0039$, recomputed in the figure's own code and asserted against the file that supplies the coefficient.

None of that resolves the singularity: the contraction along Penrose's own congruence comes out positive, so his hypothesis holds where he uses it and the theorem stands untouched. The two contractions are orthogonal combinations of the stress, and §4.2 separates them and says which one carries the sign.

The cosmological constant. One region is still unaccounted for, the interior of a cosmological horizon, and it is where a contribution to Λ would have to live. The turning is the angle a causal curve can sweep round the sphere on its way, and there it is exactly marginal. Beyond $r = L$ the turning available is $\int_L^\infty dr/(r\sqrt{|f|}) = \pi/2$ per leg, so two legs supply exactly the π the antipodal map asks for and supply it only as $r \to \infty$; at finite radius they fall short by L/r.

That marginality holds in every dimension, since $f = 1 - r^2/L^2$ does not know D, and it sits alongside the black hole's $\pi/(D-3)$, which meets the same requirement at $D = 4$ alone. Compactifying the extra dimensions does not change that: a hole larger than the compactification scale has exactly the four-dimensional turning, since motion in the compact directions only eats into the angular progress left to a causal curve, and a hole smaller than it is Tangherlini in $4+n$ supplies $2\pi/(n+1)$ against a half turn that does not move.

Adding a mass changes that. The horizon comes in to $r_c < L$, the run begins where $|f|$ is smallest, and the turning per leg becomes $\pi/2 + 2M/L$, which opens contact at $r_* = L^2/2M$. The proper time to reach that radius is $L\ln(L/2M)$, between 27 and 50 Hubble times for holes from 6.6×10^{10} solar masses down to ten. So the fold's extra term reaches nothing in the observable universe or its past. What it leaves there is the de Sitter scale $1/16\pi^2L^2$, small and nonzero, which is 2×10^{-71} of the dark energy, and no part of Λ is sourced by it. That density is hard to picture and its total is not. Over the comoving volume inside the particle horizon it sums to 4.07×10^{-17} kg, forty-one femtograms, which is 46.4 times the mass ceiling of §3.1 and more than that if the fermion is lighter. Everything the fold adds to the universe we can see weighs about what fifty of its own particles weigh. The value of Λ is a boundary datum, and the computation forces that reading.

Under the fold, then, the field equations follow from the Clausius relation, the fold and one measured constant, where they had followed from the Clausius relation, a temperature, an entropy law and a constant. What they follow as is general relativity plus one non-local term, which carries no free parameter and which no observation to date could have reached. Outside a horizon the largest value it takes anywhere is 3×10^{-34} of the dark energy.

What the fold cannot supply. The fold cannot produce a scale, and the reason is dimensional. In $\hbar = c = 1$ every structural input here carries mass dimension zero: that CPT is a symmetry of the universe, that $\Theta^2 = 1$, that the map is antilinear, that it is free at a horizon, that its transverse part is the antipodal one. It produces dimensionless numbers, eight of them: $w_0 = -1$ and $w_a = 0$ exactly, the two-body line at exactly half the mass, the exponent $2/3$ tying the decoherence time to that mass, the half-period shift that makes a horizon thermal, the equal-weight frequency $2\ln 3$, four large dimensions and no others, and the sign of G.

The fold takes two dimensionful constants, G and Λ, where one would do if the ratio between them were derivable. That ratio is a pure number, $G\Lambda \approx 3 \times 10^{-122}$, and nothing here produces it.

What the derivation does say about Λ is a statement about what kind of quantity it is. The divergence step leaves $f + \eta R/2$ constant across the whole spacetime, so Λ enters as an **integration constant** and not as a coupling. Nothing local fixes an integration constant; a boundary condition does. That is the same fact Appendix D.2 reaches from the other direction, where the restricted action gives $w_0 = -1$ and $w_a = 0$ exactly, an integration constant cannot evolve, so those are one result. It is also testable, since a robust detection of evolving dark energy would end both readings at once.

It survives a test it could have failed. If Λ were a coupling fixed by the theory's content, its rate $H_\Lambda = c\sqrt{\Lambda/3} = 1.81 \times 10^{-18}$ s^{-1} would sit at some simple ratio to a scale the fold carries. Against the dark-matter mass, the two-body line, the neutrino sum and the Planck rate, the ratios are $10^{-50.6}$, $10^{-50.3}$, $10^{-31.7}$ and $10^{-61.0}$, none commensurate with another.

Finding nothing is what a boundary datum should look like and is awkward for a coupling. So the fold does not derive Λ. It says where the value is set: the value is a fact about the fold's global state rather than about the local physics, which moves the cosmological constant problem to the bang and out of the field equations. That one number is what separates two measured constants from one.

The bang is the next place to look. The fold's condition on a cosmology is that $\Theta : \eta \to -\eta$ be an isometry of $a(\eta)^2(-d\eta^2 + dx^2)$, which needs a odd in η, and then $a'' = (K/2)\sum_i c_i(1 - 3w_i)a^{-3w_i}$ admits a fluid only where $-3w$ is an odd integer, radiation being exempt because its coefficient vanishes. Dust has $-3w = 0$ and is excluded, so a cold component at $\eta = 0$ breaks the parity. Read forwards that says the bang must be hot, which §2.2 assumed and now does not need to. For Λ the rule is permissive: $w = -1$ gives the odd power 3, so the bang admits every value and fixes none.

One route past that remains, and it closes on a number. Away from a caustic the image stress vanishes for conformally invariant matter, and the bang carries no caustic; but the Hadamard slot carrying $(\xi - 1/6)R$ carries m^2 as well, so the bang's massive content feels it. Rescaling $\eta = s/\sqrt{\gamma}$ and $p = q\sqrt{\gamma}$ removes $\gamma = M_1 a_1$ from §2.2's crossing entirely, which fixes the form to $\rho_{\rm img} = \gamma^2 H(s)/a^4$ with $s \propto a$. A cosmological constant does not depend on a, and that needs $H \propto s^4$, whereupon every factor of the scale factor cancels and the answer is cM_1^4. Against the observed $\rho_\Lambda = 2.52 \times 10^{-11}$ eV^4 that is 81.4 orders of magnitude too large.

Read the other way round, an a^{-4} density is radiation and a radiation component at the bang is what $N_{\rm eff}$ measures. The ratio to the radiation already there carries no cosmology at all. With $a_1^2/a^4 = 8\pi G\rho_r/3$ in a radiation bang, the a_1 and the a^4 cancel and $\rho_{\rm img}/\rho_r = (8\pi/3)cH(s)(M_1/m_P)^2$, about 10^{-22} at a loop factor and 10^{-20} of what $N_{\rm eff}$ can resolve.

Only the massive content contributes, since radiation is conformally invariant and the bang carries no caustic, so the scale is the fermion mass and not the bath temperature, which at the epoch $H = M_1$ would have been nine orders larger. The mass that would move $N_{\rm eff}$ by its own error bar is 7.5×10^{18} GeV, ten orders above the ceiling §3.1 derives, so the same bound that fixes the dark matter is what keeps the fold's own term out of the radiation density. So the bang cannot supply Λ either, and no part of the fold carries a scale near 2.2 meV. So Λ is a boundary datum, and no part of the fold could have set it.

Beyond it the remaining inputs are what it means to write a metric theory at all: a Lorentzian signature, and one metric carrying both the fold's horizons and a covariant matter action. Matter conservation is not among them, since Noether's second theorem gives $\nabla^a T_{ab} = 0$ from diffeomorphism invariance for any matter whatever.

What the fold adds. Jacobson's argument and the fold's version share every step from the Clausius relation to his equation (6) [33], and Appendix E marks the nine lines of general relativity that his argument already had. What differs is where the inputs come from and what comes with them. He took the temperature and the area law from the quantum field theory of the vacuum; at a flat wedge that theory's CPT operator is the fold's map, so here the inputs come from the involution that also defines the folded cosmology and fixes the dark-matter state of §2. And the fold brings what a single sheet cannot: the image stress above, a non-local term with no free parameter that acts only inside black holes, the classical-quantum split with its weight, and four large dimensions.

3.7 Commitments

Table 4 lists every observational commitment, its origin, and what a contrary observation would exclude. Most belong to the stated implementation, and several of those are inherited from [10,11] or shared with ordinary general relativity. One belongs to the structural half instead and holds whatever the implementation is: G does not run, and that follows from the same cancellation that makes the entropy coefficient universal.

Table 4. Observational commitments of the minimal implementation.

Quantity	Prediction and required assumptions	Origin	Test, and what a contrary result excludes
Light-neutrino masses	One massless light neutrino in the exact-stabilisation seesaw approximation. With normal ordering the inputs used here give $\Sigma m_\nu = 58.8$ meV and $m_{\beta\beta} = 1.5$–3.7 meV.	Refs. [10,11]; values from the stated oscillation inputs.	Oscillation measurements test the ordering; cosmology tests the sum conditional on its model. A larger established absolute mass excludes the rank-deficient mass sector. Present null double-beta results do not confirm the narrow range.
Conditional decay-energy scale	Hard $h\nu$ and $Z\nu$ energies near $M_1/2$, $E_\nu \leq 245.8 \pm 1.0$ PeV. The weak Yukawa example has $h\nu : Z\nu : W\ell = 1 : 1 : 2$ at tree level; lifetime and flavour are free; radiation and propagation determine the observed spectrum.	The abundance-to-mass machinery of [10,11] fed by the family of §2.2, plus the added Yukawa model and two-body kinematics.	Neutrino and photon spectra test a specified lifetime and flavour direction. About 46 per cent of KM3NeT's reconstructed posterior lies above the endpoint [21], so the event neither confirms nor refutes (§3.2). A channel exclusion does not exclude exact stabilisation, which predicts no decay signal.

Quantity	Prediction and required assumptions	Origin	Test, and what a contrary result excludes
Direct-recoil contact benchmark	$\sigma_n \lesssim 1.293 \times 10^{-72}$ cm^2 under the contact-rate, scattering and radiation-history assumptions; about 2.2×10^{-30} events in the LUX-ZEPLIN experiment's quoted exposure for the spin-independent point-nucleus, unit-efficiency benchmark.	Appendix D.1.	A recoil incompatible with the estimate excludes the benchmark assumptions; other $\mathbb{Z}_2$-even portals remain allowed. This row is not a live test: the cross-section sits about twenty-four orders of magnitude below current sensitivity, and compounding six generous uncertainties in the same direction closes only five of them.
Primordial tensor component from the bang	Absent in the adopted bang model without an inflationary epoch. Tensor modes obey $v'' + (k^2 - a''/a)v = 0$ for $v = ah$, and a radiation bang has $a'' = 0$, so the source vanishes at every k and not only at small k. Boyle et al. state the long-wavelength case; later radiation-era sources are untouched by this argument.	Refs. [10,11]; the all-k reading of their mechanism here.	A securely identified primordial component incompatible with that bang calculation excludes this cosmological implementation. Because a phase transition or a string network after the bang is permitted, the discriminating signature is an inflationary one: a near scale-invariant tensor spectrum across decades.
Intrinsic black-hole horizon response	Ordinary absorbing Kerr dynamics, with no added horizon reflectivity or fold-induced quasinormal-mode shift at the order treated.	The adopted minimal black-hole implementation (companion).	A robust intrinsic departure excludes that implementation; it would not identify its cause or a quotient geometry.
Newton's constant does not run	G is constant in epoch and location, with no admissible variation at any level. $\eta = 1/4G$ is fixed by a modular temperature and a transverse mode count, and §3.6 shows the surface gravity cancels out of both, so neither carries an epoch or a position.	§3.6; the same calculation that makes the entropy coefficient universal.	Any securely measured variation in G excludes the fold outright. This is the one row that tests the structural half rather than the matter sector or the implementation, and it needs no threshold, since the commitment is to no running at all rather than to a rate below a bound.
Dark energy and matter exchange	ρ_Λ constant, $p_\Lambda = -\rho_\Lambda$ and $Q = 0$ in the adopted minimal cosmology.	The restricted action analysis of Appendix D.2 and the continuity equation.	Expansion and growth data test this exact-Λ cosmology. A robust need for evolving dark energy excludes the commitment.
Growth and lensing	Standard GR baseline for the same matter content, parameters and initial spectrum in the demonstrated regime.	The retained local dynamics (Appendix A.2).	A robust departure requiring added interactions or modified gravity challenges the baseline. No all-scale completion is proved by the free algebra alone.
Quantum gravitational mediation	The graviton retains its quantum commutator; the ordinary weak-field quantum-mediation calculation remains available.	The reduced free transverse-traceless (TT) algebra (Appendix A.3); the exhibited physical quotient is one-dimensional.	A controlled gravitational-entanglement experiment tests quantum mediation. Agreement is shared with ordinary quantum gravity and does not identify the fold; the relational apparatus completion remains open.

The tensor row is the one commitment whose test is currently being argued over in public. Pulsar timing arrays have detected a nanohertz background [30], and its origin is contested between supermassive black-

hole binaries and a cosmological source. The fold does not forbid a cosmological source as such, since a phase transition or a string network after the bang is allowed. It forbids the inflationary shape, so the reading that would hurt is one in which the nanohertz signal is the low-frequency end of a near scale-invariant spectrum. The measured galaxy and black-hole population sets how much of the amplitude the binary reading can supply, and §3.4 pushes the same way: if JWST's little red dots are accreting black holes, they raise that population and the astrophysical reading gets easier. Component separation has not been done, so this is a test to watch.

The linear fold map adds no connected three-point function to the stated Gaussian free state. That is a structural result and not a complete prediction for the observable bispectrum after interactions and conversion to curvature perturbations. The fold supplies no mechanism to alter the abundance of early galaxies, an observed galaxy population is not by itself a falsifier, and early-galaxy abundances need their astrophysical and initial-condition analysis before they test the implementation. A test passed in common with GR or ΛCDM does not identify the fold.

4. Claims, limits and falsifiers

4.1 What the fold claims and where it can fail

AQFT and its thermal side fix things the cosmological accounts leave open, in the sector where the two overlap, geometry, states, horizons and correlators. The wedge-reflection theorems select the involution, the involution at the bang returns the state family, and the operator bound on that family turns the abundance into a ceiling. The influence runs both ways. Algebra contributes an input to the cosmology, the family, and not only a constraint on what the cosmology had already chosen, which is what it means to say the two legs constrain each other.

The structural result of §3.6 and the matter results of §2 and §3.1 to §3.5 share the fold and share nothing else. §3.6 uses no abundance, no dark-matter mass, no neutrino datum and no cosmological history; it uses the involution, its action at a bifurcate Killing horizon, and the geometry of a local Rindler wedge. So doubting the 491.6 PeV ceiling, or the identification of the dark matter, or the decay channel, can reject all of it and leave the derivation of the field equations standing. The dependence runs the other way too, §3.6 gives the matter sector no support it would not have without it.

Whether the derivation assumes its conclusion can be checked mechanically. Writing every derived line with what it rests on gives a graph of thirty derived nodes and fifty-seven edges over six declared roots: CPT as a symmetry of the universe, the Lorentzian signature, the one shared metric, the Hadamard condition, and the measured values of G and Λ. It is acyclic, so nothing proves itself at any distance; every leaf is one of the declared roots, so nothing enters the argument unnamed; and no path from the field equations reaches general relativity or Bekenstein-Hawking entropy, which is the loop §3.6 says is shut. Quoting the area law from Bekenstein-Hawking instead puts a forbidden ancestor in at once, and making the thermofield double depend on the field equations leaves eleven of the thirty-five nodes, roots included, unorderable, so both ways of failing are detected. The longest path from an assumption to the field equations is seven steps. One declared root appears in no edge at all, the measured value of Λ enters the cosmology and never the derivation. Tracing the field equations back reaches four of the six roots, and neither measured constant is among them: what the balance rests on is CPT, the signature, the one shared metric and the Hadamard condition. So the form of the equations is settled before either number is supplied, and G and Λ enter only where the numbers do.

One root, the Hadamard condition, is named separately, since a leaf test cannot find it. The Hadamard condition on states is what the equilibrium paragraph of §3.1 rests on and what the thermofield double uses, and a leaf test only covers what is in the graph. It is a regularity condition, the one under which a renormalised stress tensor exists at all, so it is not a property of general relativity and it does not belong in the count of nineteen against four. It is an input all the same, and it is named here.

That graph answers whether a result of general relativity was used to derive general relativity. It leaves a second question open, whether the machinery was built backwards from a destination already known. That

one is about how the work went. The work did not start from the field equations. It started from the question whether CPT taken of the universe, read through the modular structure of a wedge, is a real piece of physics or a formal coincidence. The way to find out is to ask what it is obliged to produce. If it is real, what a quantum theory of gravity has been wanted for should follow without being put in.

So §3.6 was run as a test of the premise, and the test had four places it could have failed. The surface gravity had to cancel between the occupation and the mode density. A modular temperature carrying κ spreads S/A by 21.6 and leaves no universal coefficient, so no Einstein equation. Composing the wedge reflection with the transverse antipode had to leave something frame-independent, since otherwise the balance gives one equation of state per observer and no field equation at all. The geodesic symmetry's third-order failure had to vanish with the patch, since at order unity there is no local horizon at a generic point. And the coefficient had to survive a single ultraviolet cutoff, where one tracking the surface gravity moves it by 52. Each is measured in §3.6, and none was available to adjust: every structural input carries mass dimension zero, so the fold has no free parameter with which it could have been tuned to pass.

Three things would show the fold is wrong. Their status differs.

The seam coefficient. The seam is the surface along which the two sheets join, and its coefficient says how much of a wave reaching it is reflected. If the modular structure does not fix that coefficient, the algebra leaves open a geometric quantity it ought to determine. It does not fix it. The companion adopts the transparent value rather than deriving it, and closes three routes to a derivation: one circular, one that fails whichever way the mode sum is counted, and the fold's own sheet symmetry, which the whole family of seams satisfies. What picks out the transparent point is soft and variational, the maximum of the entanglement across the seam, which locates it without excluding its neighbourhood.

One structural point survives all three closures. Seams in series compose, so the family is a one-parameter group, and the transparent value is its identity, adopting it adds no boundary term at all. That is the choice the minimal implementation makes everywhere else, in taking a linear fold map and a Gaussian free state, so a reflecting seam is the option that would need a reason. That shifts the burden of argument and is still not a derivation. The modular flow acts as the boost and is blind to any seam whose response is stationary, which covers a constant reflectivity, a frequency-dependent one and a memoryless non-linearity alike, and symmetry under the fold and closure of the constraint algebra leave the coefficient free. What survives is narrower. At quadratic order, at the specified codimension-one interface, the corner term is the only boundary term on the bifurcation surface that carries a dimensionless coefficient and still acts, so it reaches every scale-free seam. A reflecting seam remains possible, though it needs an action carrying a scale and then its own matching law. What the modular data supplies is the map, since J is what makes $\alpha = J \circ P_{\perp}$ exist, and the KMS property returns the Gibbons-Hawking temperature.

The state. If the algebra does not select the state §2.3 uses, it has failed on states, which are its own subject. This question is half open. The prescription asks two things of the state, that it be symmetric between the sheets and that it minimise the occupation. The first is forced: a Gaussian two-sheet state is Θ-invariant exactly when the two occupations are equal, mode by mode. So is the ultraviolet. The difference from the vacuum is smooth at coincidence only if $\sum_k n_k k^m$ converges for every m. A power law fails that once m passes its index, while exponential or Gaussian decay holds at every m tried. The divergence that excludes the bang-adiabatic state is the $m = 1$ case of the same sum, and the adopted state's Gaussian tail converges at every m. For a massless field the permitted power-law seed $i\gamma/4p^2$ vanishes with the mass, so a graviton occupation has no permitted power-law part at all. What remains imported is the phase $\mu(p)$ on a band of momentum, and calling the band finite is too strong, since $\mu_* + 0.1e^{-x^2}$ differs from μ_* at every finite momentum while preserving the ultraviolet expansion. That band has a description. The phase is irrelevant in the infrared, where the family's spread has closed and every μ returns $n = 1/2$; it is pinned in the ultraviolet, where a phase error δ puts a floor $\delta^2/4$ on the occupation, which makes $\int x^2 n\, dx$ and with it the relic abundance divergent for any $\delta \neq 0$, the Gaussian surviving only to $x = \sqrt{2\ln(1/\delta)}$. Exact coherence is therefore a requirement, and the state's purity at the contact supplies it. What stays free is the region between, where the phase is neither irrelevant nor protected.

No spectral quantity can close that freedom. On a fermionic mode the reduced spectrum is $\{n, 1-n\}$, and the map $n \mapsto 1-n$ carrying one end of the band to the other is particle-hole conjugation, which permutes

the spectrum and leaves it fixed as a set. Every Rényi entropy is therefore blind to the two ends, and so is any entanglement measure depending on the spectrum alone, across indices $\alpha = \frac{1}{2}, 1, 2, 3, 10$ and across the band the largest difference is 4×10^{-16}. The degeneracy is generic to any binary spectrum and is not supplied by the fold, and the residual freedom is the continuous phase, of which the two ends are only the endpoints.

Selection is dynamical. Energy is linear in the occupation, the gap between the ends is $\sqrt{1-P}$ and widens with momentum, and the upper branch is separately inadmissible. In the pairing's own variables the adopted state says that the coherence within each produced pair, $\sqrt{n(1-n)}$ for the pure pair state $\sqrt{1-n}\,|0,0\rangle + \sqrt{n}\,|1,1\rangle$, is Gaussian in momentum, their occupation satisfies $4n(1-n) = e^{-x^2}$ exactly. The scale γ is fixed by the mass the abundance determines, so what is imported is a dimensionless shape, and the shape lies inside the class the ultraviolet condition leaves without being picked out of it, since an exponential coherence with the same second moment converges at every m as well.

The band does exclude something, and the clearest case is the rival the fold itself suggests. Take the thermal occupation $n = 1/(e^{x/\tau}+1)$, the shape the fold's own map produces at a horizon. Matched to the adopted second moment of the pair coherence it needs $\tau = \sqrt{3}/\sqrt{2}\,\pi = 0.3898$, and at that scale it runs below the band's floor from the infrared out to $x = 1.81$, worst at $x = 1.11$ where it sits 31 per cent under. The band closes at $x = 0$, so what decides admissibility is the slope with which an occupation leaves one half: the floor leaves at $-1/2$ and the thermal state at $-1/4\tau$, which makes the thermal family admissible exactly for $\tau \geq 1/2$, the cubic terms leaving $1/24$ to spare there. At that boundary the production integral has the closed form $3\zeta(3)/16\pi^2 = 0.02284$, so the whole thermal family returns $M_1 \leq 389.5$ PeV and $E_\nu \leq 194.7$ PeV, which puts its line below KM3NeT's median. Θ-invariance therefore throws out the nearest alternative to the adopted state at the adopted momentum scale, and quantifies what is left of it.

A Gaussian is the least committed profile at fixed second moment, and any argument fixing the pair state variationally at fixed γ would close the question; that is a different variational principle from the minimum-energy one, acting on the profile's shape where the other acts on its level, and neither follows from the other. The residual phase is the falsifier, and its effect is bounded on one side only. Table 1 measures the excursion against a cutoff x_c, and x_c is a scale the adopted state does not have. The Hadamard tail does not remove it, because a polynomial prefactor keeps the e^{-x^2} exponent while moving the excursion's peak out to any momentum, so the production integral has no maximum and the mass no floor. What the tail does fix is the shape at infinity, and the widest excursion carrying no scale beyond the state's own, $n_{\min} + e^{-x^2}(n_{\max} - n_{\min})$, returns $I/I_{\min} = 3.75$ and $M_1 = 289.7$ PeV, which is $x_c = 1.14$ in the table. That is a benchmark, and it bounds nothing. None of the ceiling depends on any of it being closed.

The time orientations. If the two sheets' modular flows do not run oppositely, the opposite time orientation is assumed rather than supplied. Section 2.1 shows they follow from Δ.

4.2 Limits

The fold does not supply the stabilising rule. Nothing on the algebraic side refers to generations. A spacetime symmetry acts identically on every member of a flavour multiplet, so no involution on the cover distinguishes one sterile neutrino from another, whatever its charge or its lift. The rule stays an input, and a derivation would require flavour itself to be a spacetime label, which this paper neither asserts nor needs. Appendix C.2 shows that the holonomy route to the rule is closed on the cover, and that a gauged $P_\perp$ carries a mode parity where the rule needs a species parity. The lift is not a way round it either, and the lift is where a species parity would most plausibly hide. Writing the fold on a mode as $\psi \mapsto e^{i\alpha} M \psi^*(-\eta)$ gives $\Theta^2 = M\overline{M}$, in which the phase has met its own conjugate and cancelled, so a Majorana phase is no handle. And M is not free: preserving §2.2's mode equation needs $MH(\eta)M^{-1} = H(-\eta)$ with M real, which among the Pauli matrices only σ_x satisfies, giving $\Theta^2 = +1$. Change the equation so its odd part sits off-diagonal and the requirement picks σ_z instead, so the matrix is read off the dynamics rather than chosen. Every species obeying the same equation therefore carries the same lift, and the limit above is a checked statement.

Nor does the fold establish that a mirror sheet is the only possible ontology. Restrict an admissible state on the CPT extension to $\eta > 0$: the restriction of a Hadamard state to a globally hyperbolic subregion is

Hadamard, positivity survives restriction, and the correlation functions on the expanding side are unchanged by construction. A one-sided formulation supplied with those state data reproduces the production calculation exactly. What the fold does is impose a relation between the in and out descriptions and so constrain which state families are admissible at all; the bound tests that constrained preparation together with the stated history. The ultraviolet failure of the instantaneous bang vacuum is a statement about which state a one-sheet cosmology naturally reaches for, not a no-go theorem.

The algebra's causal splitting into a patch and its commutant and the cosmology's temporal splitting into branches are different splittings of one state. By direct calculation the companion shows that they cross, and nothing here identifies them.

The contact condition reaches beyond radiation. Nothing in §2.2 used a linear sweep or anything cosmological, only that H is real, the diagonal odd across the contact and the off-diagonal even; Appendix B.5 checks that across sweep laws. It does not reach the singular locus of a black hole, where the interior is Kasner and even in τ, and only the crossing type is treated here.

The interior sign is placed in this section because half of it rests on a model geometry, and the reduction that gets it there is set out in full. A stress invariant under the interior's isometries has only $T^t{}_t$, $T^r{}_r$ and $T^\theta{}_\theta = T^\varphi{}_\varphi$, and contracting with a null vector gives $|f|(k^t)^2(T^t{}_t - T^r{}_r) + r^2(k^\theta)^2(T^\theta{}_\theta - T^r{}_r)$. The fold joins a point to its image at fixed Schwarzschild t, so the contact geodesic has $E = 0$ and no t component, and its contraction is the second term alone. Penrose's congruence is radial and its contraction is the first. The two are orthogonal, so a sign computed along one does not transfer to the other, and that is the whole of why §3.6 stops where it does.

What can be said about the combination that is needed follows from conservation and the trace. With $X = T^t{}_t - T^r{}_r$, $Y = T^r{}_r$ and $A = T^\theta{}_\theta - T^r{}_r$, the trace is $X + 4Y + 2A$ and conservation is $Y' = (f'/2f)X + 2A/r$, so $X = T - 2A - 4Y$. Which term leads near the contact sphere is settled by the coupling, because the trace is $(6\xi - 1)\frac{1}{2}\Box\langle\phi^2\rangle - m^2\langle\phi^2\rangle$ and those two pieces sit on either side of the anisotropy, A carries $D^{-5/2}$, the m^2 piece $D^{-3/2}$ and the $\Box$ piece $D^{-7/2}$. At $\xi = 1/6$ the $\Box$ piece is absent, Y integrates to $D^{-3/2}$, and $X \to -2A$ with ratios of 1.056, 1.005, 1.0005 and 1.00005 as D falls through four decades, X and the anisotropy then carry opposite signs. At any other coupling the $\Box$ piece leads and X follows the trace instead, tracked to 1.00005 of it over the same range, with the contact value not reaching the leading behaviour at all.

Nothing in that is chosen. Writing the three as powers of D, conservation matches them as $c + 1 = \max(a, b)$, and the two unequal orderings each leave the trace with a single leading term whose coefficient would then have to vanish. So $a = b$ with Y one power softer is the only consistent scaling, and the trace's leading cancellation is $X + 2A = 0$. Integrating the system confirms it at four different powers, the ratio sitting at -2 to within 1.1×10^{-6} and independently of the anisotropy's size, and the power itself never enters, whatever the caustic's exponent turns out to be, the ratio is the same. The division of labour is easy to get backwards. The trace supplies the factor of two, since $X + 2A \to 0$ follows the moment Y is subleading; conservation supplies that Y is subleading. Free Y to be λA instead and the trace gives $X/A = -4\lambda - 2$, which is anything, and vanishes at $\lambda = -1/2$.

What the computed sign belongs to is then settled by a symmetry, and the answer is that the calibration geometry cannot tell the two combinations apart. On $\mathbb{R} \times S^3$ the rotations fixing a point also fix its antipode, since a rotation about an axis fixes both of its ends, so the isotropy group there is the whole of $SO(3)$ and it fixes the image point. A symmetric spatial tensor invariant under $SO(3)$ is a multiple of the identity, which the twenty-four octahedral rotations already enforce exactly, so the image stress there is isotropic and its anisotropy vanishes identically. Inside a hole the isotropy at a point is only the $SO(2)$ about its radial direction, which fixes the contact image as well but leaves the radial and transverse pressures free. Averaged over that $SO(2)$ a tensor keeps its two transverse entries equal and its radial one distinct.

So the calibration carries one combination where a hole carries two, and the combination that distinguishes them is identically zero there. Its negative value therefore says that some $\rho + p$ is negative at a hole without saying which, while conservation says that at conformal coupling exactly one of them is. What decides which is the image stress's own decomposition at the caustic, and the companion's A.18 computes it: summing the tower rather than expanding a parametrix returns every component, and the leading divergence is exactly traceless with its $T^r{}_r$ vanishing, leaving $T^t{}_t : T^r{}_r : T^\theta{}_\theta = 2 : 0 : -1$ and a coefficient $15/32\pi\sqrt{2\pi} = 0.059525$

on the offset's $-7/2$ power. Neither sign moves with the coupling or the mass, where the flat calibration's sign flips as ξ crosses $1/6$.

That pattern has a second derivation on this side of the transfer, which is stronger than the transfer would have been. Three statements force it: the leading divergence is traceless, the transverse pressures are equal by the interior's $SO(2)$, and $T^r{}_r$ vanishes at leading order, which is what $Y' = (f'/2f)X + 2A/r$ gives by putting Y one power below X and A. The trace then reads $T^t{}_t + 2T^\theta{}_\theta = 0$ and nothing is left free. So both geometries empty the same slot, for reasons with nothing in common. In the model the radial direction is not in the connecting tangent at all, and at a hole the tangent runs through it and conservation empties it anyway.

One of the two contractions carries over to a hole and the other does not, and causal character decides which. Inside a horizon $f < 0$, so r is timelike and t spacelike, and the same three components mean different things. The contact geodesic has $E = f\dot{t} = 0$ with f nonzero inside, so its tangent is a timelike direction plus a transverse one, which is what the contact direction is in the model geometry as well. In the model the radial direction is the spacelike one, and its transverse sphere has constant radius, so it carries no tidal field there. The contact contraction therefore carries over, and it is negative: the anisotropy at a hole is negative, conservation puts $X = -2A$ above zero, and $T_{ab}k^a k^b$ along Penrose's ingoing radial congruence is positive, so the null convergence condition holds there.

Carrying a sign between two geometries is the step to distrust, so the companion settles it on the one quantity that can be computed both ways. The image term at a caustic follows from A.19's rule, which replaces the divergent Van Vleck factor, the measure of how geodesics from one point spread, by an arc length and a Gaussian, and it follows independently from A.18's mode sum, which carries no Van Vleck factor at all. Applied to A.18's own geometry the rule has nothing left to choose, because the non-degenerate transverse direction there is flat and its reduced factor is exactly one, and it returns the mode sum's coefficient to sixteen digits. So the image term is the same positive expression in both, for the same reason, a reduced Van Vleck factor is a square root, an arc length is positive. A residual phase between the two would have to be a multiple of π, since a Wightman function at spacelike separation is real, and a multiple of π would flip a sign that is the same in both.

The two halves come out with opposite signs. The image stress defocuses the congruence that would bring the two sheets into contact, delaying the conjugate point the contact requires, and focuses the one Penrose's argument runs on. So the fold censors its own closed causal curves and leaves the singularity theorem's hypothesis where it found it. What is still held by conservation rather than by a sum done in the interior is the radial sign itself, and the same sum on Schwarzschild would remove that step. Its hardest ingredient is in hand: the interior's zero-frequency radial modes are Jacobi polynomials, $r^2 P_{\ell-1}^{(2,0)}(1-2r)$ with $2M = 1$, whose phase is $2\nu \arcsin\sqrt{r} - 5\pi/4$ with the variable part equal to the companion's own contact leg, from r out to the horizon.

The magnitude does not depend on any of this. The shell thickness rests on the coefficient and the power, and neither cares which combination carries the sign.

One limit applies to the whole of §3.6. What a Clausius argument returns is an equation of state. The field equations hold at every point and in every null direction, and no gravitational action is produced along the way. So nothing here supplies a path integral, a graviton propagator, or a route to quantising gravity, and §3.6 shows the field equations do not need one, not that one has been found. The image term carries the same point, since it is non-local and a local gravitational action would not have produced it in any case.

The remaining mathematical tasks are specific: the interacting constraints, the joint matter-and-apparatus construction, and a cover matter action and measure that enforce the sterile-species rule from the proposed boundary data. A general rotating-horizon seam needs its own matching law. None of the present results requires those objects to be silently assumed complete.

4.3 Two questions the fold raises

The fold's two-sided horizon results need a hole with a past singularity to be identified with, which a hole formed by collapse does not have, so for an astrophysical black hole they say nothing. That divides black holes into two classes, and the division is not observational, a transparent seam predicts ordinary Kerr dynamics either way. If the overmassive early population of §3.4 needs seeds that did not form by collapse, those seeds are the only objects in this framework that could be two-sided, and there is room for them with nine orders to spare. We have no test that separates the classes and do not propose one; which black holes have a past is left open and named.

The fold reverses charge as well as time, so the mirror sheet carries the opposite sign of every conserved charge this one does, and Boyle et al. already read the cosmological baryon asymmetry that way [10]. In the two-level reduction the two components a mode shares equally at the contact are its positive- and negative-frequency amplitudes, so $|\psi_1(0)| = |\psi_2(0)| = 1/\sqrt{2}$ reads as exact particle-antiparticle balance at the bang, and the band's two ends are charge conjugates that no spectral measure tells apart. The identification of the components with particle and antiparticle content is a reading of the reduction, the freedom between the endpoints is a continuous phase and not a two-element choice, and saying more needs the interacting theory this paper does not have.

4.4 What the fold takes and what it returns

A claim to unify is easy if the model is allowed to overfit. Here is the parameter count. The rule used throughout is that a quantity is free if the theory permits a range and observation picks a point in it, and fixed if the theory returns a number that was not chosen. Postulates are counted apart from parameters, since removing one is a different kind of gain.

Take the baseline to be general relativity with ΛCDM, a dark-matter particle and massive neutrinos. In the sector this paper touches, six things are free and settled by fitting: Newton's constant, the value of the cosmological constant, w_0 and w_a, the dark-matter particle mass, and the neutrino mass scale. The fold returns four of them. The dark-matter mass is 491.6 ± 2.0 PeV, a ceiling the abundance saturates. The neutrino mass scale is $\Sigma m_\nu = 58.8$ meV, from one massless light neutrino. And $w_0 = -1$ with $w_a = 0$ exactly, from the restricted action of Appendix D.2. What remains taken from experiment is Newton's constant and the value of Λ, which is two dimensionful constants, the same two any formulation of gravity needs.

Counted as prior volume the dark-matter row dominates. The literature searches that mass from fuzzy dark matter near 10^{-22} eV to the GUT scale, fifty decades, and the fold returns it to four parts in a thousand. The neutrino row spans fewer decades, about half of one, and more in sharpness: the prediction is that the sum sits exactly on the oscillation floor, at the edge of the allowed band, and one established absolute mass ends it. And w_0, w_a collapse from a plane to a point.

The other half of the count is postulates. The divide between classical and quantum behaviour comes out, because $\Theta^2 = 1$ makes the sheet average the fold-even part and the difference the fold-odd part, which obey the algebra of the Keldysh classical and quantum variables on the half-period contour of §2.1. So do the Unruh temperature and the field equations, though neither was ever put in by hand: quantum field theory gave the first [37] and Jacobson derived the second from it [33]. What the fold changes for those two is their source, the one supposition that CPT is a symmetry of the universe itself, not just of the laws inside it.

Counted line by line, nineteen separate properties of general relativity come out and four go in; Appendix E is the list. Nine of the nineteen were already in Jacobson's argument and four are standard results. Six need the folded universe: the black-hole involution, the contact surfaces, four dimensions, the classical-quantum split, the sheets' opposite time orientation and the hot bang. Each of the nineteen names the file in the release that produces it. Three of the four name the file that examines them rather than one that derives them, and the fourth is the value of G, which no script can supply. The script that produces the list stops without printing if a named file is absent. Those four are a Lorentzian signature, one metric carrying both the fold's horizons and a covariant matter action, and the values of G and Λ. The first two are what writing

a metric theory means; the last two are the dimensionful constants no symmetry argument can supply, since every structural input here carries mass dimension zero.

Two entries one might expect are absent, and in each case what the argument uses is narrower than the name. The equivalence principle is not assumed. Its weak form is what the eikonal limit of any field on the shared metric does, the characteristics of the wave equation agreeing with the Christoffel system to 2×10^{-12} with the mass cancelling out of the path, and a matter action built from a different metric sends the rays along that metric's geodesics instead. Diffeomorphism invariance is not assumed either, because its two uses have different status. The contracted Bianchi identity, which is what leaves exactly one integration constant, is a theorem about the Levi-Civita connection of any metric: measured on a generic curved metric it sits at 2×10^{-5} of the Einstein tensor's own size and falls as h^2, while a connection with torsion breaks it at order unity. The covariance of the matter action, which is what $\nabla^a T_{ab} = 0$ comes from, is part of the metric postulate already listed, since T_{ab} means the variation of that action against that metric and has no other meaning. What is left on the assumed side is nothing substantive about gravity. Figure 5 draws the whole count.

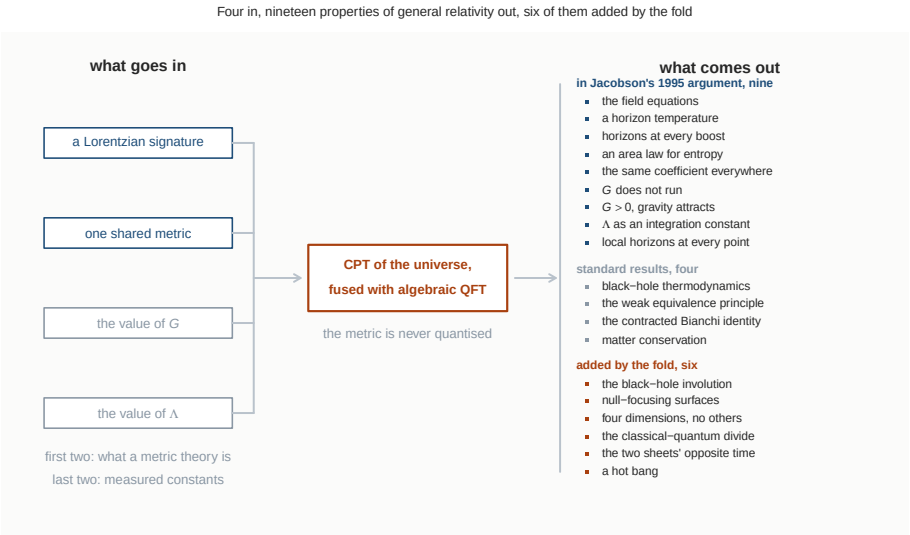


Figure 5. What goes in and what comes out. On the left is everything the fold is given: a Lorentzian signature and one metric carrying both the fold's horizons and a covariant matter action, which between them are what writing a metric theory means, and the measured values of G and Λ. In the middle is the single postulate, that CPT is a symmetry of the universe, fused with the algebraic quantum field theory that fixes the map. On the right are the nineteen properties of general relativity that follow, the metric staying a fixed background throughout, grouped by where each stood before: nine in Jacobson's 1995 argument, four standard, and six that need the folded universe. The dependency graph behind the two columns is acyclic and no path from the field equations reaches general relativity or Bekenstein-Hawking entropy, which is what stops the argument assuming its own conclusion. The labels are the list's own lines and their counts are read from it, so the figure cannot drift from the text, and Appendix E gives the same lines at full wording. What the figure does not draw, because it is not a property of general relativity, is the Hadamard condition on states, which §4.1 names as a root of the dependency graph.

The signature entry deserves more than a line saying the fold is built on a spacetime that has one, because the fold is not indifferent to it. Everything the fold takes from Θ comes from its being a discrete symmetry with a parity, and a parity is invariant only if the map cannot be deformed continuously to the identity. In four Euclidean dimensions it can: rotating by θ in the 1-2 plane and by θ in the 3-4 plane at once is an isometry at every θ and reaches $-\mathrm{Id}$ at $\theta = \pi$, so the fold would be a rotation, removable, with no fold-even and fold-odd sectors to speak of. In Lorentzian signature it cannot, because every $\Lambda \in O(1,3)$ satisfies $|\Lambda^0{}_0| \geq 1$, confirmed over twenty thousand sampled boosts and rotations with a smallest value of 1.0000001 against a largest of 4969, while the same entry in $SO(4)$ passes freely through zero. A continuous path of isometries cannot carry $+1$ to -1, so $-\mathrm{Id}$ is PT and sits in a component of its own. The check does not

refuse everything, a spatial rotation by π, which has $\Lambda^0{}_0 = +1$, is reached by an explicit path.

That comparison is two-way and the question is not. A four-dimensional manifold can carry $(0,4)$, $(1,3)$, $(2,2)$, $(3,1)$ or $(4,0)$, and the count runs over all five. $O(t,s)$ has four components labelled by the signs of the determinants of its timelike and spacelike blocks, neither sign changes along a path of isometries, and $-\mathrm{Id}$ carries $(-1)^t$ and $(-1)^s$. So it sits in the identity component exactly when t and s are both even. Pairing like-signature directions and rotating both pairs through π builds the path explicitly at $(0,4)$, $(2,2)$ and $(4,0)$, and at $(1,3)$ and $(3,1)$ there is no path of any kind: three thousand sampled elements of each identity component keep both block determinants positive, while $-\mathrm{Id}$ has one negative. What separates the two cases is the difference between a rotation and a boost. Like-signature directions rotate into each other and reach -1 on both at π; a timelike and a spacelike direction only boost, and the boost's diagonal entry is cosh, which is never below one. An unpaired time direction has nothing to pair with.

So the fold keeps a parity exactly when the number of time directions is odd, which in four dimensions leaves $(1,3)$ and $(3,1)$, one signature written two ways. None of this says a Lorentzian manifold has to exist. It says that among the arenas one could have written the fold on, it has content in exactly one, which is a different kind of assumption from a convenience. The dimension selection of §5.2 in the companion and this one are not independent, since a causal comparison already presumes a signature to be causal in, and together they say only that the pair (4, one time direction) is the one the fold can live on.

Two coefficients are open, the seam coefficient and the state's residual phase $\mu(p)$, and §4.1 assesses both. They are structural and are fitted to nothing, so they are not free parameters in the sense counted here. The entropy coefficient η is likewise not an extra input, since $\eta = 1/4G$ and a theory of gravity cannot avoid one dimensionful constant.

The count, then, is this. One symmetry assumption goes in. Four numbers that used to be fitted come out, the dark-matter mass alone narrowing from fifty decades to four parts in a thousand, and so does the classical-quantum divide, while Jacobson's two inputs gain a single source. Nothing is added in their place. That is the sense in which this paper claims to unify.

4.5 What would refute it, and what no test can take away

One event securely assigned to the decay component above 245.8 ± 1.0 PeV refutes the fixed-history implementation, and about a hundred assigned events give a calibrated population test with power 0.88 against an endpoint twice as high. A cosmological bound on Σm_ν that tightens by more than 8.4 per cent under ΛCDM excludes the exact-stabilisation sector, and the model cannot move to evolving dark energy to escape it. A compact-object fraction above one per cent of the dark matter moves the line by more than its width, and a primordial seed population nine orders denser than observed would leave none for the sterile neutrino. A near scale-invariant primordial tensor spectrum excludes the bang model.

The two dimensionful constants are both frozen, and that is a single commitment. Λ is an integration constant of the derivation, so it does not run by construction, and the fold cannot add a running piece because its own extra term reaches 2×10^{-71} of the dark energy where the universe has been observed. η is fixed by a modular temperature and a transverse mode count, neither of which depends on epoch or location, so G cannot run either. The fold takes two numbers from experiment and holds both fixed. A secure measurement of either varying ends it, with nothing available to absorb the result. The seam coefficient and the state's residual phase are the two things the algebra leaves undetermined, and both are stated as open.

One structural consequence does not depend on any of those tests. Since $\Theta^2 = 1$, the average of a field's two sheet readings is exactly its fold-even part and the difference exactly its fold-odd part, so the fold's parity is the classical-quantum split itself: where the sheets agree the difference vanishes and the physics is classical, and where they disagree is where the commutator lives. Nothing is laid over the Keldysh algebra to get that. It is what the algebra says once the second contour leg is a place.

The amplitude of the quantum half is then not free either. Fold-invariance forces $\langle \Theta A \Theta \rangle = \langle A \rangle$ for Hermitian A, so where Θ exchanges two things their expectations are equal, which is why Section 3.1's contact condition is a consequence; and where Θ fixes the object, the identity says nothing. At a horizon the selection

arrives from the other direction and arrives complete, because the fold's map being the half-period shift makes the equilibrium cross-sheet correlator the direct one shifted by $i\beta/2$, hence a thermofield double with $\tanh r = e^{-\beta\omega/2}$. That same squeeze is the crossing's ceiling $\operatorname{arcsinh}(1/\sqrt{2})$, at $\beta\omega = \ln 3$. A second relation comes out of the same matrix. Its eigenvectors are the fold-even and fold-odd combinations, with eigenvalues $\coth(\beta\omega/4)$ and $\tanh(\beta\omega/4)$ in units of $1/2\omega$, and their product is 1 at every frequency and every temperature, so the geometric mean of the classical and quantum weights is the zero-temperature amplitude. Nothing in the normalisations above disturbs that, since $\langle\Phi_c\Phi_c\rangle$ is half the first eigenvalue and $\langle\Phi_q\Phi_q\rangle$ is twice the second, and the two halves carry equal weight at $\beta\omega = 2\ln 3$. The two ends of the fold meet at one frequency, and the transverse parity is carried through because the map has no fixed points.

Those are the commitments, and they are what observation has to be set against.

5. Conclusions

The structural half of this paper was not what the work set out to find. Suppose the mirror symmetry behind every particle experiment, the one that swaps matter for antimatter and runs time backwards, is a symmetry of the universe itself and not just of the physics going on inside it. Three things usually expected of a quantum theory of gravity then come out without anything gravitational being quantised.

Where the classical world comes from. The two mirror copies of a field have an average and a difference, and those are exactly the classical and quantum variables physics already uses, so the divide between them is the mirror's own parity and not a scale that anything has to cross. Why a horizon is hot: the mirror's map at a horizon is a shift by half the thermal period, which makes the correlation between the copies a thermal one. That is the Unruh and Hawking temperature reached from the mirror's side, and no temperature was put in anywhere. Those two turn out to be one fact. The weight the quantum half carries is $4\tanh^2(\beta\omega/4)$, so it fades out for modes softer than the horizon's own temperature, and the classical world is a horizon seen from below.

Third, the field equations themselves. Jacobson's 1995 argument gets Einstein's equations out of thermodynamics, given a horizon temperature and an entropy proportional to horizon area, and he took both from the quantum field theory of the vacuum. That theory has the mirror in it already: at a horizon its CPT symmetry is the mirror's map, and the same map makes the horizon hot and its entropy an area. So the argument runs on the symmetry that splits the classical from the quantum, with nothing taken from relativity in order to prove relativity. Of the nineteen properties of general relativity counted in §4.4, nine were in Jacobson's argument already and four are standard; the mirror adds six. What comes out is listed in §4.4; what goes in is two statements about what a spacetime theory is and the two numbers, Newton's constant and the cosmological constant, that every theory of gravity has to be told. Four large dimensions are on the output side of that list. The signature and the single shared metric are inputs, and §4.4 shows the fold has content only with one time direction. The metric is a fixed background on every page of this paper.

Now the matter half, where two ideas that had never been set side by side do what neither does alone. If the mirror is a symmetry of the universe itself, the Big Bang has a far side, a whole mirror universe running away from it the other way, and asking which quantum states that mirror permits makes a number fall out. The mirror forces every ripple in the dark-matter field to sit evenly between the two states open to it, and a ripple split evenly cannot come out empty, since empty would mean all of it sitting in one state. So the bang has to make particles, and there is a floor under how many. How much dark matter there is has been measured, so fewer particles means each one is heavier, and the floor turns into a ceiling. Dark matter can weigh no more than 491.6 ± 2.0 PeV, about half a billion proton masses. No fit produced that number; it is what the symmetry allows.

Breaking the model is easy. The dark-matter particle decays in two, so it should leave a sharp neutrino line at half its mass, 245.8 ± 1.0 PeV, and one well-measured neutrino above that line ends it. It also forces the lightest neutrino to weigh nothing at all, which fixes the three neutrino masses at 58.78 ± 0.32 meV between them, and it has given up the usual way of wriggling out of that. Two more things come free. A black hole dating from the bang has two outsides and can feed from both at once, and since each side only feels the glare of its own infalling matter, the hole grows at roughly twice the rate either side thinks possible, the

head start JWST's surprisingly heavy early black holes look like they need. And on one further assumption the fold has no room for a master time the universe as a whole could be running in, though the everyday kind still ticks for anyone inside it.

The field equations do not come out as general relativity exactly. Running the balance on the fold's own horizons puts the image stress into it, and that term is not a local curvature polynomial, so it cannot be absorbed into Newton's constant and this is not general relativity with a shifted coupling. It carries no free parameter: away from the contact sphere its size is set by the matter's departure from conformal invariance and by nothing else. Outside a horizon a point and its image are spacelike separated, which gives not a zero but a suppression, exponential in the mass for anything heavy, quadratic for anything light, Weyl-suppressed for a massless conformal field, with the worst case anywhere in that range thirty-four orders below the dark energy. So the fold diverges from relativity only where relativity already predicts its own breakdown, on the inner half of a black hole's interior, and agrees with it everywhere else.

That divergence has a size. The single region where general relativity predicts its own breakdown is the single region where the fold leaves it, and the departure runs in the direction that keeps the sheets from touching. The region is a shell around the contact sphere whose thickness follows from the order of the caustic there. At a hole of one solar mass, half a femtometre for an electron, a seventieth of a nanometre for a top quark, and two microns for the dark-matter particle the rest of this paper is about. It widens as the hole does, so the place to look is the largest holes. Nothing in the coefficient is fitted, and the order-unity factor that used to stand beside it is measured at one.

The shell does not thin away to nothing as the mass falls, which is the last thing the calculation had to say. At the contact sphere the coupling cancels out of the leading term, so a massless conformal field feels it too, and what is left is a floor at $\ell_P^{4/7} r_h^{3/7}$ up to a number: 1.1×10^{-19} m at a solar mass, sixteen orders above the Planck length, with no matter content in it at all. Nothing the fold can be made of falls through that.

The departure does not touch the singularity, and that is computed here. The fold joins a point to its image at the same Schwarzschild time, so the direction in which the two sheets touch carries no time component, while the congruence Penrose's argument follows is the radial one. Conservation and the trace tie the two contractions together at conformal coupling with a minus sign, so exactly one of them is negative, and the companion's A.18 computes the whole tensor at the caustic and says which. The contact contraction comes out negative. The null convergence condition therefore fails along the direction in which the sheets would touch and holds along the one Penrose uses: the fold defocuses its own closed causal curves and leaves his theorem where it found it.

Here the metric is a spectator, so none of this shows that a quantum theory of it does not exist or is not needed elsewhere. The entropy coefficient is not part of that limit: it is $1/4G$, of mass dimension two, and every structural input of the fold has mass dimension zero, so no argument from a symmetry could ever have returned it. What a symmetry returns is relations and dimensionless numbers, and this one returns both, none of it finished. The fold does not say why one neutrino species is stable, does not prove a mirror sheet is the only way to read it, and leaves two coefficients open, each quantified in §4.2.

One result is left for last. Einstein's equations came out of a symmetry, with the metric a fixed background at every step and no gravitational field ever promoted to an operator. Neither constant's measured value enters that derivation, as §4.1's dependency graph shows by tracing the balance back to four of its six roots; G and Λ are needed where the numbers are and nowhere before. Read that way, general relativity is what the two-sheet construction looks like from one sheet. This derivation of the field equations did not need a quantum theory of gravity. Whether the metric admits a quantum theory of its own is a separate question, and this paper leaves it open.

Appendix A. The fold's algebra

A.1 The closed-time-path weight and the kernel dictionary. With the variables of §2.1 the action weight is

$$\exp\Big\{ i \Big[S\Big(\Phi_c + \frac{\Phi_q}{2}\Big) - S\Big(\Phi_c - \frac{\Phi_q}{2}\Big) \Big] \Big\},$$

the relative sign following from exchanging the legs while conjugating i. One reading has to be excluded because it would make the weight empty. If Φ_c and Φ_q are built from a single history and its literal image under Θ, then for a real matter action on a Θ-invariant domain $S[\Phi \circ \alpha] = S[\Phi]$ by change of variables, the exponent vanishes identically, and no dynamics survives. That is a correct statement about histories and not what is meant. The relation imposed is between the two observable algebras. An antiunitary map relating left and right actions does not force their difference to vanish. On Hilbert-Schmidt space with $JX = X^\dagger$ one has $JL_H J = R_H$ for Hermitian H while $(L_H - R_H)X = [H, X]$ is not the zero operator, and the resulting evolution is ordinary. The weight is read with Φ_c and Φ_q as independent integration variables carrying the algebraic relation of §2.1. That dictionary fixes the free two-patch sign and the variables; it does not establish a nonlinear measure, a general interacting state or a probability for choosing a sheet, and it does not identify expectation prescriptions without their state, ordering and boundary data. The Gaussian marginal of A.2 is an expectation in the positive cover state; an unshifted trace contour and a symmetric thermal contour retain their own boundary conditions.

In the thermal construction the legs may be displaced by half the KMS period, as in the symmetric contour of Niemi and Semenoff [18] and Herzog and Son [19]. In the stated de Sitter quasifree state the geometric and contour cross-leg kernels are

$$G_{12}^{\mathrm{fold}}(x, y) = W(x, \alpha y), \qquad G_{12}^{\beta/2}(x, y) = W(x, Jy), \qquad \alpha = J \circ P_\perp .$$

On the central worldline $P_\perp$ acts trivially and the two kernels agree. Away from it the fold supplies a transverse parity, $(-1)^\ell$ mode by mode. A closed-time-path calculation can incorporate that twist; the claim is that the stated geometry supplies it, not that it lies outside the formalism. In the conformal-scalar control of the companion, $Z_\alpha - Z_J = -2r^2$. Take both points to the horizon. The conformal scalar's kernel is $G \propto 1/(1-Z)$; J leaves the transverse direction alone while α carries it to the antipode, so $1 - Z_J \to 1 - \cos\gamma$ and $1 - Z_\alpha \to 1 + \cos\gamma$, and

$$\frac{G_\alpha}{G_J} \longrightarrow \frac{1 - \cos\gamma}{1 + \cos\gamma} = \tan^2 \frac{\gamma}{2}.$$

The two agree at $\gamma = \pi/2$, where a point and an antipode are equidistant; the thermal kernel is singular at $\gamma = 0$, where the points coincide, and the fold's at $\gamma = \pi$, where a point meets the antipode. Each is the ordinary short-distance singularity of a two-point function at the angle its own identification selects. The closed form belongs to the de Sitter conformal scalar and its maximal symmetry; at a general bifurcate Killing horizon only those three features survive, and a near-horizon argument carrying the closed form to Schwarzschild does not hold (companion, A.10).

A.2 The fixed real algebra and the local dynamics. The Θ-fixed real algebra is generated by the even fields Φ_c and by $i\Phi_q$. Their mutual commutator is nonzero, its complexification reconstructs the generated quantum algebra, and projection onto Φ_c alone discards the conjugate sector. With the normalisation of §2.1 the even covariance is $\langle \Phi_c \Phi_c \rangle = (G_{\mathrm{sym}} + G_{\mathrm{img}})/2$, and the commuting even sector has a Gaussian probability distribution given by the Bunch-Davies vacuum marginal in this polarisation. A covariance is not a probability density and this marginal is not a complete classical dynamical theory; the statement concerns the identified global algebra and does not make every local algebra on every quotient spacetime classical.

The paper carries two invariant sets and only one of them is a complex algebra. Conjugation by the antilinear Θ is antilinear on the algebra, so its fixed set is closed under real combinations but not under multiplication by i: it is a real von Neumann algebra, Takesaki's theorem does not apply to it, and no unital complex-linear expectation onto it exists, since $E(i\mathbb{1}) = iE(\mathbb{1}) = i\mathbb{1}$ while $i\mathbb{1}$ is not Θ-fixed. The subalgebra fixed by the linear $P_\perp$ is complex. Transverse parity commutes with the boost, since they act on different factors, checked

in static coordinates and in the $\mathbb{R}^{1,4}$ embedding where the boost mixes (X_0, X_1) and $P_\perp$ is $-\mathbb{1}$ on (X_2, X_3, X_4). The $P_\perp$-fixed subalgebra is therefore invariant under the modular flow, so by Takesaki's theorem a normal state-preserving conditional expectation onto it exists and its modular operator is the restriction of the full algebra's. The fold does not disturb the modular structure: the horizon's thermodynamic data is what it was, the local dynamics can be retained rather than assumed, and no entropy built from the modular operator can depend on a seam coefficient.

Unchanged local dynamics has explicit premises. On a common globally hyperbolic, causally convex patch, with the same locally covariant time ordering and renormalisation prescription, compactly supported interactions and insertions, and equality of the full smooth restricted two-point function, naturality makes the relative S-matrices and Bogoliubov maps agree coefficient by coefficient [4,5]. Equality of Hadamard singularities alone is insufficient. The supporting results are Hackl and Neiman's, that the antipodally even solution space is Lagrangian and each observer's algebra is the ordinary static-patch algebra [4], and Friedman and Higuchi's restricted atlas [5]. Global state preparation, noncompact support, gauge and edge sectors, and sums over saddles remain outside this local statement, so no unconditional or all-orders local invisibility is asserted.

A.3 A nonzero group-averaged free-graviton sector. Harlow and Numasawa [25] argue that a spacetime inversion cannot be a global symmetry of quantum gravity and must be gauged; averaging over the constraints is that gauging. Their setting, closed universes and the Euclidean path integral, differs from the free linearised field on de Sitter treated here, so what follows is not their computation, and admitting a non-orientable manifold into a sum over topologies is not the same as identifying spacetime with the quotient, since the sum contains the cover as well.

For group averaging we use the standard global, temporal-gauge radiative TT Fock representation of Marolf and Morrison [13], importing its one-particle norm, invariant vacuum and compensated de Sitter action together. Gérard and Wrochna [14] separately construct a positive $O(4)$-invariant Hadamard state whose modified covariance is not invariant under the full de Sitter group; we do not identify that state with the representation used here.

Choose real spatial TT harmonics T_{nA}, $n \geq 3$, whose spatial antipodal parity on either TT type is $s_n = (-1)^{n+1}$. For future-increasing conformal time and the positive Klein-Gordon convention, the canonically rescaled positive-frequency mode is

$$u_n(\eta) = \frac{e^{-in\eta}(\tan\eta - in)}{\sqrt{2n(n^2-1)}}, \qquad u_n(-\eta) = -\overline{u_n(\eta)},$$

with $i(\overline{u}_n u_n' - u_n \overline{u}_n') = 1$; an even scale factor converts back to the metric field without changing the reflection identity. On one-particle coefficients define $(\Theta_1 c)_{nA} = (-1)^n \overline{c_{nA}}$, an antiunitary involution. Its bosonic Fock lift fixes the vacuum, sends $a^\dagger_{nA}$ to $(-1)^n a^\dagger_{nA}$ and implements $\Theta h(p)\Theta^{-1} = h(\alpha p)$ with tensor pullback understood. Temporal gauge and the TT conditions are preserved, and an isometry maps pure-gauge perturbations to pure-gauge perturbations, so the map is well defined on the reduced classes. Since the antipode is central, the lift and the compensated de Sitter action implement the same automorphism of the irreducibly represented reduced canonical-commutation-relation algebra, their scalar ambiguity fixed to one by the common invariant vacuum. Hence $\Theta U(g)\Theta^{-1} = U(g)$.

The $O(4)$-invariant two-graviton seed $\Psi = \sum_{A=1}^{10} (a^\dagger_{3A})^2 |0\rangle$ is the unique parity-even quadratic creation-operator singlet in the real $n = 3$ transverse-traceless basis, with $\|\Psi\|^2 = 20$ and $\Theta\Psi = \Psi$. It is not already physical, the six rotation charges annihilate it while the four boost charges obey $\sum_{a=1}^{4} \|Q_a \Psi\|^2 = 240$. The full $SO_0(1,4)$ average is therefore required. In the spin-2 boost chain the adjacent-level coefficient is $b_n = \frac{1}{2}\sqrt{(n+3)(n-2)}$ and the bottom matrix coefficient is $d(\lambda) = \operatorname{sech}^6(\lambda/2)$; the KAK decomposition and $O(4)$ invariance give $\langle \Psi, U(k_1 b(\lambda) k_2)\Psi\rangle = 20 \operatorname{sech}^{12}(\lambda/2)$, so the integral reduces to one rapidity variable. With the Haar factor of [13],

$$\langle \Psi, \eta(\Psi)\rangle = \|\Psi\|^2 \int_0^\infty \sinh^3\lambda \, \operatorname{sech}^{12}(\lambda/2)\, d\lambda = 20\left(\frac{2}{3}\right) = \frac{40}{3},$$

the substitution $u = \tanh(\lambda/2)$ giving $16\int_0^1 u^3(1-u^2)^2 du = 2/3$. Convergence alone does not prove positivity on an unspecified finite-particle domain, so we use the explicit cyclic domain $\mathcal{D}_\Psi = \mathrm{span}_{\mathrm{alg}}\{U(g)\Psi : g \in SO_0(1,4)\}$. For $v = \sum_i c_i U(g_i)\Psi$ and $w = \sum_j d_j U(h_j)\Psi$, invariance of Haar measure gives

$$B(v,w) = \int_{SO_0(1,4)} dg\, \langle v, U(g)w\rangle = \frac{40}{3}\overline{\sum_i c_i}\sum_j d_j,$$

so B is positive semidefinite on the whole domain, its null space is $\sum_i c_i = 0$, and its completed quotient is one-dimensional with $B(\Psi,\Psi) = 40/3$. The formula is independent of the chosen finite orbit expansion. If $\sum_i c_i U(g_i)\Psi = 0$, pairing that relation with Ψ under the same absolutely convergent average gives $\sum_i c_i = 0$.

The one-dimensionality is not the result. Group averaging the cyclic orbit of any seed gives a one-dimensional quotient, since $\eta U(g)\Psi = \eta\Psi$ for every g; that is a property of Haar averaging and carries no information about the fold. What is fold-specific is the step that could have failed and did not. The averaging imposes the gravitational constraints, and an antiunitary has no general right to survive one. It survives here because $\Theta U(g)\Theta^{-1} = U(g)$, which needed the scalar ambiguity fixed to unity by the common invariant vacuum, and because $\Theta\Psi = \Psi$ makes $\mathcal{D}_\Psi$ a Θ-stable domain. Given those, $B(\Theta v, \Theta w) = \overline{B(v,w)}$ and the lift descends to conjugation on the physical sector. So the antilinear structure and the free gravitational constraints are compatible, which is the claim at issue; it is a claim about Θ and not about the dimension of the quotient. Had Θ failed to commute with the de Sitter action, the fold would not have survived constraint imposition at all.

This establishes neither positivity on the full finite-particle space, nor a change to the Keldysh algebra, nor an interacting gravitational theory, and a one-dimensional sector carries no local observables, so nothing here is a physical prediction. The reduced free transverse-traceless algebra retains its ordinary graviton commutator, so the standard weak-field quantum-mediation calculation, including the Bose-Marletto-Vedral proposal, remains available [15]. The result does not select a unique graviton vacuum: fold covariance leaves a curve in each Bogoliubov disc, and the Euclidean cap supplies the additional state choice. Nor does it complete the interacting gravitational constraints or the relational apparatus sector.

A.4 The fold requires a constraint formulation, given one premise. One requirement follows from the fold's antilinearity together with a premise stated rather than derived, that physical vectors are exactly Θ-invariant at every external time. Ordinary CPT covariance, $\Theta\psi(t) = \psi(-t)$, relates complete histories, is consistent with evolution in an external time, and does not on its own force $H\psi = 0$.

The requirement is elementary. Θ is antilinear, so if the dynamics is CPT-covariant, $\Theta H\Theta^{-1} = H$, antilinearity flips the sign in the exponent and $\Theta\, e^{-iHt}\,\Theta^{-1} = e^{+iHt}$. A Θ-invariant state therefore evolves to $\Theta|\psi(t)\rangle = |\psi(-t)\rangle$ and stops being Θ-invariant: the identity holds exactly at four dimensions tested, while a Θ-invariant state acquires a discrepancy of order one within $t \sim 1$. Energy eigenstates retain invariance only up to a phase and a generic superposition loses it outright. A state solving a constraint $H\psi = 0$ carries no phase either, and its invariance is preserved exactly, to 7×10^{-16} at $t = 0.9$ and 8×10^{-16} at $t = 5$. So if the fold requires physical states to satisfy $\Theta|\psi\rangle = |\psi\rangle$, the formulation cannot carry an external time evolution and the state must solve a constraint. Three results in the companion depend on it. The reality of the wavefunction, and with it equal branch weights, obtained from Θ-invariance rather than from the no-boundary proposal; the coincidence of the singularity condition's sector with the Θ-invariant one, which holds exactly when there is no evolution between the branches; and the self-adjointness of the projector, which holds only in the same limit, the obliquity running from under a degree at evolution parameter 0.01 to 82° at 2. Nothing in §3 depends on them.

What is established is the conditional, and the antecedent is a choice. Appendix A.3 gauges the linear involution $P_\perp$, following Harlow and Numasawa; gauging an antilinear map is a different operation, which we do not perform, and we adopt the antecedent. That other operation carries consequences. An antiunitary involution with $\Theta^2 = 1$ defines a real structure: its invariant vectors form a real Hilbert space $H_\mathbb{R}$ with $H = H_\mathbb{R}\otimes\mathbb{C}$, of half the real dimension, which we confirm directly. To gauge Θ is to say the theory is real. For any Hermitian A with $\Theta A\Theta^{-1} = -A$, antiunitarity gives $\langle A\rangle = -\langle A\rangle^*$ and hence $\langle A\rangle = 0$ exactly in

every gauged state, while Θ-even observables are unconstrained; tested on three thousand random Hermitian operators of each parity, identically zero for the odd ones and order unity for the even.

Since Θ is CPT the odd observables are the CPT-odd ones, baryon number among them, so a Θ-gauged theory carries zero net baryon number across the pair while energy, being CPT-even, does not cancel. That is the CPT theorem's ordinary consequence; what the fold adds is the reading of the pair as two physical sheets, which turns a selection rule into a statement about where the balancing matter is. Harlow and Numasawa do gauge an antiunitary inversion, which is what produces the real Hilbert space above. What differs is this paper's choice: A.3 gauges the linear $P_\perp$ and carries the antilinear map as a real structure on the algebra. The twisted sectors are then open. A gauged linear $\mathbb{Z}_2$ has sectors classified by $H^1(M;\mathbb{Z}_2)$, which Appendix C.2 shows vanishes on the cover, while a gauging of the antilinear map is not classified by that cohomology and we have not worked out what classifies it.

A.5 Complete histories and the return charge. Linearisation stability on a closed spatial slice requires the de Sitter charges of a complete perturbation to vanish. For a fold-invariant complete stress history the antipode reverses the future cover normal, so $Q_\xi(-t) = -Q_\xi(t)$, while stress-tensor conservation and Stokes' theorem give $Q_\xi(-t) = Q_\xi(t)$. All ten cover charges therefore vanish, and in this limited sense the image supplies the return the complete-history constraint requires. The restriction is essential: the antipode maps $(t, \mathbf{n})$ to $(-t, -\mathbf{n})$, so the image lies on the same slice only at $t = 0$. A finite branch preparation still needs an apparatus Ward flux, and the joint matter-plus-gravity rigged norm and the relational laboratory observable are open. The result balances a complete history and does not remove the apparatus from a finite experiment.

Appendix B. States at the bang

B.1 The excluded state. Two states are easy to mistake for the adopted one. The "in" vacuum's occupation is $|\beta|^2 = e^{-x^2}$, giving $I_b = \pi^{-2}\int x^2 e^{-x^2} dx = 1/(4\pi^{3/2}) = 0.0448968$. That is not the integral §2.3 uses. The adopted half-angle state gives $I = 0.0127597$, smaller by a factor 3.52, and since $M_1 \propto I^{-2/5}$ the two differ by a factor 0.605 in the mass. Θ-invariance settles it before any integral is done. At zero momentum the sweep is perfectly diabatic, so the in vacuum's occupation there is 1, where the fold forces exactly 1/2 and the band has zero width, the miss is the whole of the available range. It re-enters the band at an algebraic point, since $e^{-x^2} = n_{\max}$ gives $4u^2 - 3u = 0$ with $u = e^{-x^2}$, so $u = 3/4$ and $x = \sqrt{\ln(4/3)} = 0.5364$, outside below and inside above. The comparison of integrals records the difference; the symmetry is what excludes the state.

The bang-adiabatic vacuum starts its evolution at $\tau = 0$, so first-order adiabatic perturbation theory retains a boundary term. With $\theta = \arctan(p/\gamma\tau)$ the adiabatic coupling is $\theta'/2$, and integration by parts leaves $\frac{1}{2}\cdot\theta'(0)/(2i\omega(0)) = i\gamma/(4p^2)$, whence $n_B \to \gamma^2/(16p^4)$. Numerically $n_B/(\gamma^2/16p^4)$ runs from 1.114 to 1.0010 at $p/\sqrt{\gamma} = 2\ldots 6$, with log-slope from -4.308 to -4.006. A p^{-4} tail leaves $\int dp\, p^2 n(p)$ convergent, while the relativistic energy measure $\int dp\, p^2\omega_p n(p) \sim \int dp\, p^3 n(p)$ is logarithmically divergent. This is the mode-occupation statement, not a covariant point-split stress tensor.

The failure is a bulk divergence, $\langle T_{\mu\nu}\rangle = (m^2H^2/16\pi^2)\ln\Lambda\ \mathrm{diag}(1, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, traceless, conserved, $\propto a^{-4}$ and non-zero at every $\eta > 0$, a divergent amount of radiation created at the bang and redshifting through the interior. A boundary counterterm's stress tensor is $\delta(t)$-supported and $\delta(t) \neq \theta(t)f(t)$, so no boundary term absorbs it. The log coefficient is reproduced at $\eta = 0.5, 1, 2, 5$ as 0.006330 against the predicted $\gamma^2/16\pi^2 = 0.0063326$. That obstruction is specific to $w = 1/3$, the sudden-start amplitude $\propto M'(0)$ vanishing for dust, and §3.6's parity rule reaches the same place from the other side: Θ admits a fluid only where $-3w$ is an odd integer, which excludes dust and leaves radiation as the case the bang saturates.

The Hadamard condition on a fermion mode at a radiation bang fixes the ultraviolet expansion, whose leading seed ratio is $c_+(0)/c_-(0) = i\gamma/4p^2 = i(ma_1/4)(-\nabla^2)^{-1}$, nonlocal in the spatial Laplacian. Nadal-Gisbert, Navarro-Salas and Pla parametrise CPT-invariant fermion states with this requirement and obtain further Hadamard candidates by a low-energy selection [22]. Boyle, Finn and Turok's half-angle vacuum has the required leading seed and cancels the sudden-start amplitude ($\theta_0 \to \pi/2 - \gamma/4p^2$, $n(0) \to \gamma^2/16p^4$, verified to six to nine digits). The asymptotic condition does not eliminate smooth finite-momentum changes, so

the ultraviolet test excludes the sudden-start alternative and does not select a unique member of the CPT-invariant Hadamard family.

B.2 The ultraviolet criterion acts on the phase function. Because μ_* moves with p, what Θ leaves is a phase function, so the reduction is from a magnitude and a phase per mode to a phase alone, and the ultraviolet criterion acts on that function. A phase held constant in p leaves the occupation tending to a constant, with log-slope falling only from -0.56 to -0.20 across $p \in [1, 3]$. The bang-adiabatic value $\mu = \pi$ gives log-slopes -3.9 to -4.4, the p^{-4} tail whose energy integral diverges. Only $\mu_*(p)$ cancels the boundary term down to the Gaussian: the minimum over μ sits at a ratio $0.2524, 0.2513, 0.2519, 0.2321$ to $e^{-\pi p^2}$ at $p = 1.00, 1.25, 1.50, 1.75$, the factor $1/4$ the half-angle relation predicts. Past $p \simeq 2.25$ the ratio is numerical noise: the cancellation needed is 6×10^{-10} at $p = 3$ and 10^{-18} at $p = 4$, below what double precision holds, so the range ends where the integrator does and not where the claim does. Within that range Θ-invariance and Hadamard regularity between them fix the ultraviolet behaviour of the state with no appeal to minimum energy.

B.3 Dephasing does not give one half. A bang is a violent environment, and a dephased two-level system loses its coherent transition probability, so $n = 1/2$ might be an environmental artefact. The two strong-dephasing limits go to opposite ends. Lindblad dephasing added to the swept Hamiltonian in the diabatic basis freezes the populations, the quantum Zeno effect, and drives $n \to 0$. At $\Delta = 0.4$ it falls $0.225 \to 0.181 \to 0.122 \to 0.045$ as Γ runs $0, 10, 50, 200$ in units of the sweep rate. In the adiabatic basis the same dephasing enforces following instead and drives $n \to 1$, reaching 0.96 at $\Gamma = 50$. One half is the single value neither limit reaches, and it is not approached from either side, and the coherent limit reproduces $1 - e^{-\pi\Delta^2/2\gamma}$ to the truncation error of the sweep. This rules out the easy confound; it does not show that no environment produces one half by some other route, and the dephasing rate is phenomenological rather than derived from the cosmological environment.

B.4 Kramers, and the sign of the exchange. An antiunitary CPT on fermions often squares to $(-1)^F$, and Kramers' theorem would then forbid invariant states in the odd sector. In the two-mode Fock space built explicitly, with $\Theta c_1 \Theta^{-1} = c_2$ and $\Theta c_2 \Theta^{-1} = -c_1$, the map applied twice sends $c_i \mapsto -c_i$, so $\Theta^2 = (-1)^F$ exactly. The odd sector is Kramers-degenerate, the pair states this paper uses are even, $\Theta^2 = +1$ there, and no obstruction arises for them. Nor is a Kramers degeneracy established by the square alone: that needs the relevant Hamiltonian to commute with the antiunitary, which the history-reversing mode map does not by itself supply.

The sign of the exchange is not forced. With $\Theta c_2 \Theta^{-1} = -c_1$ the invariant paired state is $\sqrt{1-n}\,|00\rangle + \sqrt{n}\,|11\rangle$ at every n; with the opposite sign it is $\sqrt{1-n}\,|00\rangle + i\sqrt{n}\,|11\rangle$, also at every n, and both produce, so requiring production selects neither. The two full lifts are not related by a phase convention: a unitary change of basis sends an antiunitary square to a unitarily conjugate square and cannot turn fermion parity into the identity. They differ on the odd sector, where the minus-exchange lift squares to $(-1)^F$ and the plus-exchange lift to $+1$, and this paper takes nothing from either. What a phase convention can relate is the invariant even-sector pair amplitude.

B.5 How far the contact condition reaches. Nothing in §2.2 used radiation, a linear sweep or anything cosmological, only that H is real, that the diagonal is odd across the contact and that the off-diagonal is even. For a crossing written as $D(\tau) = A\,\mathrm{sgn}(\tau)|\tau|^n$, the identity $H(-\tau) = \sigma_x H(\tau)\sigma_x$ holds exactly for every n, at machine zero across $n = 0.25$ to 4, so the conclusion is insensitive to the sweep law and does not depend on the bang being radiation dominated. Unitarity supplies the complement: $|a_1|^2 + |a_2|^2 = 1$ makes the family symmetric about a half, $n_{\max} = 1 - n_{\min}$, with phase average exactly $1/2$, both holding to nine figures at sweep exponents $n = 0.5$, 1 and 2. So at a crossing of this type Θ fixes the magnitudes and unitarity fixes the phase-averaged occupation, neither consulting the sweep law; what the geometry sets is the transmission through the contact. $n = 1$, the radiation value, is the single exponent at which the crossing is the exactly solvable Landau-Zener problem, and it is the exponent B.1 argues is forced at the bang on separate grounds. We have no closed form at other exponents, and a single stretched-exponential fit is the wrong model there, since the transmission develops interference structure in p.

The condition does not follow at the singular locus of a black hole, and we do not claim it. The oddness is the whole load, supplied at the bang by $a(\eta) = a_1\eta$ changing sign through the contact. A Schwarzschild

interior is Kasner with $r \propto |\tau|^{2/3}$, which is even, so writing $\mathrm{sgn}(\tau)$ into D there would put in the hypothesis. The companion gives that locus a different structure. The two singular branches are exchanged rather than crossed, and the consistency requirement is the global fixed-point equation $(JU)x = x$ on data traversing the interior, not a local involution at a surface a mode passes through. Whatever contact a fold of this kind makes, the condition it imposes there is fixed by the involution and not by the metric.

Appendix C. The quotient readings

C.1 The projected prescription at one loop. On S^4 of radius R the conformal propagator is $1/(4\pi^2 d^2)$ in chordal distance and the antipodal chordal separation is $d = 2R$, so $G(x, Ax) = 1/(16\pi^2 R^2)$, obtained also as an Abel-summed spectral series with partial sums 1.08033, 1.00755, 1.00075, 1.000075; the agreement checks the implementation of a fixed identity and is not an independent proof. With the cover's Hadamard subtraction this leaves $\Delta\langle\phi^2\rangle = \eta/(16\pi^2 R^2)$ and, for $\lambda\phi^4/4!$, the 1PI tadpole $\Delta\Sigma = \eta\lambda/(32\pi^2 R^2)$. The projected prescription is not the physical observer state: a detector-response calculation rejects its literal Lorentzian kernel, and the image term is neither the two-point function of any state on the cover algebra [5] nor what a patch observer measures [4].

C.2 The $\mathbb{Z}_2$ line bundle. These are limitations of the quotient reading, which is the natural first proposal. *(i)* Both real flat line bundles are allowed and geometry does not choose which species tensors with the non-trivial one; calling it a Wilson line presupposes a $\mathbb{Z}_2$ gauge bundle, whereas a non-trivial fundamental group supplies possible holonomies, not a dynamical gauge field. *(ii)* A descended spinor in sign sector ε obeys $\psi(Ax) = \varepsilon J_A \psi(x)$, whence $\psi = (\varepsilon J_A)^2\psi = J_A^2\psi$: the holonomy changes the equivariance eigenvalue and cannot change the square of the spacetime lift. *(iii)* No fermion determinant for the full Standard Model plus three neutrinos has been constructed on this non-orientable Lorentzian quotient. The valid conditional statement is that a chosen compatible Pin^+ lift together with a chosen non-trivial real line bundle for one species forbids odd operators in that species at the quotient-bundle level.

The physical cover has $H^1(S^3 \times \mathbb{R}; \mathbb{Z}_2) = 0$ and the quotient bundle pulls back trivially. Because the cover is simply connected its H^1 vanishes for every coefficient group, so the two-sheet reading admits no holonomy sector, no bundle choice and no discrete angle to fix; the choice in *(i)* is a freedom the quotient has and the fold does not, which leaves the quotient needing a parameter geometry does not supply. That negative is about holonomy only.

A $\mathbb{Z}_2$ can also arise as the matter representation of a gauged symmetry, and Harlow and Numasawa's argument that a spacetime inversion must be gauged [25] supplies one. The fold has a linear candidate: Θ and the antiunitary modular conjugation are both antilinear, so their composition is linear and implements $P_\perp$, an involution with charge $(-1)^\ell$. To gauge it one must assign each matter field a representation, and that assignment is what a species rule is. This does not derive the stabilising rule, because the charge is a mode parity and the rule is a species parity, and nothing forces the sterile sector into the odd representation. The proposed Euclidean boundary condition has not been continued into a Lorentzian cover matter action and measure that preserve the same odd-species selection rule, so the stabilising $\mathbb{Z}_2$ of [10,11] is retained as an additional assumption and the particle is not called stable without qualification. Production happens at conformal time $\tau_* \simeq 1/\sqrt{\gamma}$ while the future boundary at which a horizon identification would act is at $\eta_* = 4.39/H_0$, giving $\tau_*/\eta_* = 1.6 \times 10^{-25}$, and the largest conceivable holonomy-sector differential in n_{dm}/s is 5.8×10^{-26}.

C.3 Two corrections to the literature-facing reading, and the polarisation on the quotient. *(i)* Friedman and Higuchi's (3.22) is not an image state. At the two-point level their antipodally symmetric object reads $\tilde{W}(x,y) = \frac{1}{2}[W_0(x,y) + \overline{W_0(Ax,Ay)}] = \frac{1}{2}[W_0(x,y) + W_0(Ay,Ax)]$, a state on the cover's global algebra containing no $G(x,Ay)$ anywhere and constraining a single-patch observer's correlators not at all. It is not a state on any global algebra of the quotient, because no such algebra exists (Kay's F-locality no-go); its content on the quotient is the collection of restrictions to a lifted atlas, which is why their extra atlas condition, that the closure of $U \cup V$ be time-orientable for every pair, is needed. Their own positivity failure, $\rho_1'(IO) = 2c(c-1)/(1+c^2)$, is negative for $0 < c < 1$ with minimum -0.4142135624 at $c = \sqrt{2} - 1$, and is a positivity failure under the prescribed gluing of the correlations rather than a singularity of the

antipodal image (a 12-level Fock truncation agrees with the closed form to 2.2×10^{-16}). *(ii)* Dulac and Wei's no-boundary density matrix on elliptic de Sitter [6] is neither a restriction nor an image state. Section 5 of their paper identifies creation and annihilation operators and obtains a one-dimensional global Hilbert space; the quotient "does not admit a global time orientation". Their construction contains the counterpart of A.2's commutator theorem in its variables, while their observer Fock space uses static-patch operators that are not A-even, and the two are never joined (§6 of their paper). *(iii)* The polarisation does not descend. In the normalisation of §2.1, $[\Phi_c, \Phi_q] = i\Delta$ with $\Delta \circ (A \times A) = -\Delta$, so the sign of the canonical commutation relation is not single-valued on a non-time-orientable quotient, and the A-even symplectic current is a twisted 3-form on RP^3 (four fundamental domains of one involution give -0.63, -2.64, -4.17, $+3.18$). Elliptic dS_4 is non-orientable, A reverses total orientation and time orientation and preserves spatial orientation.

Appendix D. Benchmarks and the exact-Λ derivation

D.1 A contact-rate and scattering benchmark. The direct-detection estimate concerns one heavy-contact benchmark and not a derived ultraviolet operator. Its order-of-magnitude rates are

$$\Gamma_{\rm int} \sim \frac{T^5}{\Lambda^4}, \qquad H(T) = 1.66\sqrt{g_*}\frac{T^2}{M_{\rm Pl}}, \qquad \sigma_n \sim \frac{m_n^2}{\pi\Lambda^4},$$

with $g_* = 106.75$ and the non-reduced Planck mass. If this effective description remains valid at bath temperature $T = M_1$, requiring the interaction rate there to lie below the expansion rate gives $\Lambda \gtrsim 9.588 \times 10^{10}$ GeV and $\sigma_n \lesssim 1.293 \times 10^{-72}$ cm^2. The coefficients and form-factor treatment are the conventions of this benchmark, so these are order-of-magnitude estimates, not portal-independent limits, and the single-temperature check does not establish nonthermalisation throughout the earlier radiation era. With the standard radiation inputs, the epoch $H = M_1$ has bath temperature about 1.870×10^{13} GeV, where the ultraviolet portal must be specified.

For the isospin-conserving spin-independent contact benchmark the zero-momentum nuclear cross section is $\sigma_A(0) = A^2(\mu_A/\mu_n)^2\sigma_n$, with μ_A and μ_n the dark-matter reduced masses with the nucleus and nucleon, and nuclei rather than nucleons are counted in converting flux to events. At $M_1 = 4.916 \times 10^8$ GeV, $\sigma_n = 10^{-72}$ cm^2 and $A = 131$, a local density of 0.4 GeV cm^{-3} and mean speed 230 km s^{-1} give approximately 2.2×10^{-30} events in LZ's 2.84 tonne-year exposure with $F(q) = 1$ and unit efficiency; nuclear form factors, recoil thresholds and efficiencies reduce a physical count. One event at this idealised rate corresponds to about 0.015 Earth masses of xenon operating for 13.8 billion years at today's assumed halo flux, or 0.012 Earth masses at the calculated contact ceiling. These are exposure-scale comparisons and not detector designs. Exact $\mathbb{Z}_2$ permits other even portals, so a recoil incompatible with this estimate excludes this implementation of the coupling and not the particle, the massless-neutrino result or the two-copy construction.

D.2 Exact Λ from the restricted action. For the restricted metric-only action class used in the companion derivation, with expansion scalar $\theta = 3H$,

$$\rho = \theta f'(\theta) - f(\theta), \qquad \rho + p = -\dot\theta f''(\theta).$$

The stated off-shell closure conditions leave a constant term plus a boundary term. For the constant term $p_\Lambda = -\rho_\Lambda$ even in a decelerating matter-filled background, and the continuity equations then give $Q = 0$ and $\rho_m \propto a^{-3}$. This is a result for that restricted action and closure problem, not a theorem about every covariant dark-energy theory. The local GR equations are unchanged in the sectors demonstrated here, so for the same cosmological parameters and initial spectrum the baseline uses the standard growth and lensing relations; any correction to the Poisson equation would have to be specified as part of a different effective implementation and included in its predictions. A linear transformation of a Gaussian free state preserves its vanishing connected three-point function, so the fold adds no bispectrum in that calculation; a full primordial prediction also needs the interaction terms, initial state and conversion to the observable curvature perturbation.

Appendix E. Inputs and outputs, line by line

§3.6 counts nineteen properties of general relativity out and four in, and Figure 5 draws the count with the lines compressed to labels. Here they are at full wording, grouped by where each stood before this paper. Nine were already in Jacobson's 1995 argument [33], which took its temperature from the Unruh effect [37], its area law from vacuum entanglement with a fixed cutoff as in Bombelli, Koul, Lee and Sorkin [38], and its local horizons from local flatness. They are re-derived here from the fold's map, which at a flat wedge is the CPT operator itself (§3.6), so what is new in them is where they come from and not that they hold. Four are standard results of general relativity and of quantum field theory in curved spacetime. Six need the folded universe itself. Each derived line names, in the release, the file that produces it, and the checking pass fails if this list and that file's list come apart or put a line in different groups.

E.1 Nineteen that come out.

Already in Jacobson's 1995 argument.

1. **The form of the field equations,** $R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab}$ proportional to T_{ab}. Jacobson's equation (6) [33]. Clausius run on the fold's own horizons, with Raychaudhuri integrated, the null-vector algebra checked, and the Bianchi step written out.

2. **A temperature at every local Rindler horizon.** Jacobson took it from the Unruh effect [37]. Here the fold's map is the half-period shift, so the cross-sheet correlator is a thermofield double; by Bisognano-Wichmann [28] that is the same statement.

3. **Those horizons, at every boost and orientation.** Jacobson had them from the local Poincaré symmetry at any point. Here the wedge reflection and the transverse antipode are each frame-dependent and their composition $-\mathrm{Id}$ is not.

4. **An entropy proportional to the horizon area.** Jacobson motivated it from the entanglement of vacuum fluctuations across the horizon, with a cutoff, as Bombelli and coauthors and Srednicki computed it [38,39]. Here it is the entanglement entropy of the state the fold forces, with flat-space transverse mode counting. Neither argument uses Bekenstein-Hawking.

5. **The same entropy coefficient at every horizon,** which is what makes the result Einstein's rather than one equation of state per observer. Jacobson's single cutoff length gives it, and Jacobson's later work names a finite, universal coefficient as the input still assumed [40]. Here the surface gravity cancels between the occupation and the WKB mode density, because the wedge metric in boost coordinates carries none and Bisognano-Wichmann fixes the modular temperature at 2π. S/A holds to 10^{-9} while β runs over four decades, and the cancellation survives three different transverse weightings. The condition is the one Jacobson's cutoff already carried: it must be a single proper length, since one tracking κ moves S/A by a factor of fifty.

6. **That Newton's constant does not run** with epoch or location. Immediate from a single coefficient, as in Jacobson's $G = 1/4\hbar\eta$. Here η is fixed by the modular temperature and a transverse mode count, neither of them cosmological.

7. **The sign of Newton's constant,** so that gravity attracts. $8\pi G = 2\pi/\eta$ and η is an entropy density, which is positive; the same holds in Jacobson's argument.

8. Λ **appearing at all, as an integration constant.** Jacobson's equation (6) has it in the same place. The contracted Bianchi identity and matter conservation leave exactly one constant of integration, and both of those are derived below, at lines 11 and 12.

9. **Local Rindler horizons away from the fold's own fixed locus.** Jacobson's local flatness. Here the geodesic symmetry at any point has the fold's differential $-\mathrm{Id}$ and is an isometry through second order, which is the order the balance is computed at. Its failure is third order in ∇R and its weight against the balance vanishes linearly in the patch size.

Standard results.

10. **Black-hole thermodynamics.** Hawking [42] and Israel [7]; Sewell derived the Hawking temperature from PCT [2], and Kay and Wald proved the thermal state unique at a bifurcate Killing horizon [41]. The fold reads it as the same half-period shift as line 2.

11. **The contracted Bianchi identity,** which is what leaves exactly one integration constant. A theorem about the Levi-Civita connection of any metric, measured at 2.2×10^{-5} relative on a generic curved metric and falling as h^2; a connection with torsion breaks it at order unity.

12. **Matter conservation,** $\nabla^a T_{ab} = 0$. Noether's second theorem applied to a covariant matter action, which Jacobson assumed outright. The divergence of the stress tensor is the matter field equation contracted with the field, checked off shell in a generic curved metric.

13. **Matter following the metric's geodesics,** the weak equivalence principle. The eikonal characteristics of a field on that metric are its geodesics, checked against the Christoffel system to 2.4×10^{-12}, with the mass cancelling out of the path.

Added by the fold.

14. **The transverse involution $P_\perp$ at a black hole.** The only free involutive isometry of a round bifurcation sphere is $-\mathrm{Id}$, and of Kerr's axisymmetric surface the antipodal map; the uniqueness theorems make those the whole stationary vacuum case.

15. **Null-focusing surfaces for Raychaudhuri to act on.** Contact is conjugacy, so the fold's own contact surfaces are caustics, at every charge and in every dimension.

16. **Four large spacetime dimensions and no others.** The turning available for contact closes only at $D = 4$: a half turn is required because the antipode is an involution of a sphere, and the interior supplies $\pi/(D-3)$ per leg, which at $D = 5$ reaches it only in a limit that is no point of the spacetime and from $D = 6$ falls short outright. What is derived is that four is the only dimension in which the sheets touch anywhere, so a higher-dimensional fold is allowed and has nothing to add.

17. **The classical-quantum divide, and its amplitude.** The variables of Schwinger and Keldysh are standard [16,17], and so is the symmetric thermal contour of Niemi and Semenoff [18] on which the odd part's weight against the even one comes out as $4\tanh^2(\beta\omega/4)$. What is the fold's is that the two parts are its even and odd sectors, so the divide is a parity of the universe.

18. **The opposite time orientation of the two sheets,** which the cosmological accounts assume. The modular flow of a wedge is the boost, whose generator gives $dX_0/ds = X_1$, positive at every point of the right static patch and negative at every point of its antipode. That flow is Bisognano and Wichmann's; what the fold adds is that the antipode is the other sheet. Two limits hold with it: the flow preserves the state, so no arrow of time is derived, and the relation doing the work is $\Delta_{A'} = \Delta_A^{-1}$ between an algebra and its commutant rather than any property of J.

19. **A hot bang, with no cold component at the fold's fixed slice.** Θ invariance needs a odd in conformal time, which admits a fluid only where $-3w$ is an odd integer and excludes dust, so the bang must be hot. §2.2 assumed that and no longer needs to.

E.2 Four that go in.

1. **The value of Newton's constant.** Dimensionally impossible for a symmetry to supply: every structural input of the fold has mass dimension zero and $\eta = 1/4G$ has dimension two.

2. **The value of Λ.** An integration constant of the derivation, and both places it could have come from are closed. The turning available is short at every finite radius in the late universe, and at the bang an a-independent image stress would be of order M_1^4, some 81.4 orders above the observed value.

3. **Lorentzian signature.** The fold is built on a spacetime that already has one, and no other signature gives the fold content: $-\mathrm{Id}$ sits in the identity component of $O(t,s)$ exactly when t and s are both even, so the fold keeps a parity exactly when the number of time directions is odd. Over the five four-dimensional signatures that leaves $(1,3)$ and $(3,1)$ and excludes Euclidean, $(2,2)$ and $(4,0)$ alike.

4. **One metric, carrying both the fold's horizons and the matter action.** The Clausius step needs it already, since T_{ab} means the variation of the matter action with respect to that metric and has no other meaning. Not a convenience either: with a second metric, conservation holds against the wrong connection and the mismatch is 4×10^5 times the numerical floor, while a constant rescaling, which changes no connection, leaves 7×10^{-15}.

Two of the four are what writing a metric theory means and two are units. Two entries that used to sit on this side have moved across. The equivalence principle went, because its weak form is what the eikonal limit of a field on the shared metric does, and diffeomorphism invariance went because its two uses have different status. The contracted Bianchi identity is a theorem about any metric, and the covariance of the matter action is part of the metric postulate rather than beside it. Special relativity is input rather than output, and nothing here bears on the gauge structure of matter; the derivation starts from both.

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Code and data availability

Every quantitative claim in this paper is reproduced by a script in the release. There are 197 of them, 24903 lines in total: 175 calculations and 22 figure generators, each naming in its header what it computes and

what it does not settle. 5 of those files carry a banner at both ends saying the result they compute has been superseded, and print it when run, with the reason kept in the file. 19 of the generators and 149 of the calculations are R, and none of them loads a package, so base R is enough; they run to completion under R 4.5.2, one taking several minutes. The remaining 26 calculations and 3 generators are Python, run under 3.12. 7 use only the standard library and the rest import numpy, scipy, sympy or mpmath, at the versions the release records. A checking pass runs 28 gates over both manuscripts and fails if a quoted number has drifted from the script that produces it, if a cross-reference points at a section that does not exist, if a figure is absent or unused, if a label runs off the edge of its panel, or if a position stated in one place is contradicted in another. Each gate carries a planted case it has to catch, because a check that matches nothing reports success.

The release accompanying this version, with the manuscript source, the figure assets and drawing code, those scripts and a SHA-256 manifest, is at https://github.com/BenWiseman/separate-ways (tag v5.6), archived at https://doi.org/10.5281/zenodo.23034297, the concept DOI of the code deposit, which covers every release. It is not the complete private development history. The exploratory dark-energy fits and Kerr-barrier calculations have separate Python sources and dependencies and are not among the R scripts counted above, and Appendix C.1's antipodal series is checked in the companion's release rather than this one, since that is where the rest of the antipodal-propagator work sits. Pantheon+ data are external inputs, with acquisition instructions, provenance and exact hashes in `data/README.md`. The manuscript has a separate Zenodo record at https://doi.org/10.5281/zenodo.22888119, again a concept DOI, which covers every version and always resolves to the most recent; each individual deposit also carries its own version DOI, and that is the one to cite when a specific set of numbers matters. The companion paper, *Somewhere Over the Horizon*, has its own Zenodo record at https://doi.org/10.5281/zenodo.23030633, also a concept DOI.

Passing these checks shows that the stated formulas and inputs give the quoted numbers. It does not test the physical assumptions behind them. The scripts and the checks over them were written with the help of the language models named in the acknowledgements. That is the reason for the discipline above: every script states in its header what it settles and what it does not, and every check carries a case it has to fail on.